

(MR20-1CS0306) DATA ANALYTICS WITH PYTHON

III Year B. Tech – II Semester

Dr.M.Narayanan

Professor / CSE

COURSE OBJECTIVES:

- ❖ Learning about the Importance of Data and its importance
- ❖ Knowing Python fundamentals and Pandas essentials
- ❖ Learning the Principles of Probability and sampling Methods
- ❖ Getting knowledge about formulating and testing hypothesis
- ❖ Learning and analytical comparison's with ANOVA methods
- ❖ Learning about Performance indicators using ROC methods

UNIT-IV

UNIT-IV

Analysis of Variance (ANOVA): Introduction to ANOVA, one way ANOVA, two way ANOVA, Post – Hoc test

Regression: Simple Linear Regression, Multiple Linear Regression, Maximum Likelihood Estimation (MLE), Logistic Regression, step-wise methods and algorithms

Reference Books:

1. McKinney, W. (2012). Python for data analysis: Data wrangling with Pandas, NumPy, and IPython. " O'Reilly Media, Inc."
2. Swaroop, C. H. (2003). A Byte of Python. Python Tutorial.
3. Ken Black, sixth Editing. Business Statistics for Contemporary Decision Making. “John Wiley & Sons, Inc”.
4. Anderson Sweeney Williams (2011). Statistics for Business and Economics. “Cengage Learning”.

Introduction to ANOVA

- ❖ Analysis of Variance (ANOVA) is a statistical method used to compare the means of three or more groups to determine if there are significant differences among them. It helps in identifying whether variations in the data are due to actual differences between groups or just random chance.
- ❖ **ANOVA compares the variation between group means to the variation within the groups.**
- ❖ Analysis of Variance, is a test used to determine differences between research results from three or more unrelated samples or groups.
- ❖ Types of ANOVA include **One-way (for comparing means of groups)** and **Two-way (for investigating effects of two independent variables on a dependent variable).**

Why Use ANOVA?

- ❖ When comparing only two groups, a t-test is sufficient.
- ❖ However, for three or more groups, multiple t-tests increase the risk of Type I error (false positives).
- ❖ ANOVA solves this by testing all group differences in one test.

When to Use ANOVA:

- ❖ **If the difference between the two populations is statistically significant, then the two populations are unequal.**
- ❖ Now that we have understood what ANOVA is, let's understand some important terms related to ANOVA.

Important Terms Related to ANOVA

Means (Grand and Sample):

- ❖ A sample mean is the average value for a group, whereas the grand mean is the average of sample means from various groups or the mean of all observations combined.
- ❖ The grand mean is the overall mean of all data points from all groups combined

F-Statistics:

- ❖ F-statistic or F-ratio is a statistical measure that tells us about the extent of difference between the means of different samples. Lower the F-ratio, closer are the sample means

Formula for F-Statistic

$$F = \frac{\text{Between-Group Variance}}{\text{Within-Group Variance}}$$

Sum of Squares

- ❖ The sum of squares is a technique used in regression analysis to determine the dispersion of data points. It is used in the ANOVA test to compute the value of F.

Mean Squared Error (MSE)

- ❖ The Mean Squared Error gives us the average error in the data set.

Hypothesis

- ❖ In ANOVA, we have Null Hypothesis and an Alternative Hypothesis. The Null hypothesis is valid when all the sample means are equal, or they don't have any major difference.
- ❖ The Alternate Hypothesis is valid when at least one of the sample means is different from the other.

Group Variability

- ❖ In ANOVA, a group is a set of samples within the independent variable.
- ❖ **Between-group variability** occurs when there is a significant variation in the sample distributions of individual groups.
- ❖ **Within-group variability** occurs when there are variations in the sample distribution within a single group.

One-Way ANOVA

- ❖ **What is one-way ANOVA?**
- ❖ The most common method of performing an ANOVA test is one-way ANOVA. The one-way ANOVA means that the analysis of variance has **one independent variable**.
- ❖ One-way analysis of variance (ANOVA) is a statistical method for testing for differences in the means of three or more groups.
- ❖ **How is one-way ANOVA used?**
- ❖ One-way ANOVA is typically used when you have a single independent variable, or factor, and your goal is to investigate if variations, or different levels of that factor have a measurable effect on a dependent variable.
- ❖ **Use the one-way ANOVA to see if there are any significant differences between the means of independent variables.**
- ❖ **When you know how each independent variable's mean differs from the others, you can figure out which of them is linked to your dependent variable and start to figure out what's driving that behavior.**

What are some limitations to consider?

- ❖ One-way ANOVA can only be used when investigating a single factor and a single dependent variable.
- ❖ When comparing the means of three or more groups, it can tell us if at least one pair of means is significantly different, but it can't tell us which pair. Also, it requires that the dependent variable be normally distributed in each of the groups and that the variability within groups is similar across groups.

One-way ANOVA is a test for differences in group means:

- ❖ One-way ANOVA is a statistical method to test the null hypothesis (H_0) that three or more population means are equal vs. the alternative hypothesis (H_a) that at least one mean is different. Using the formal notation of statistical hypotheses, for k means we write:

$$H_0: \mu_1 = \mu_2 = \dots = \mu_k$$

H_a : not all means are equal

where μ_i is the mean of the i -th level of the factor

- ❖ Performs a **One-Way ANOVA (Analysis of Variance)** test to determine if there is a significant difference in test scores among three different teaching methods (Method A, Method B, and Method C). The goal is to perform a One-Way ANOVA to test if there is a significant difference in the mean test scores between the three methods. Data:
- ❖ **Method A: 85, 88, 90, 92, 87**
- ❖ **Method B: 78, 80, 82, 85, 81**
- ❖ **Method C: 88, 85, 84, 87, 90**
- ❖ **Let's use the example of three teaching methods (Method A, Method B, and Method C) with the following test scores:**

Method A: 85, 88, 90, 92, 87

Method B: 78, 80, 82, 85, 81

Method C: 88, 85, 84, 87, 90

Step 1: Set Hypotheses

- ❖ Null Hypothesis (H_0): There is no significant difference in the mean test scores between Method A, Method B, and Method C.
- ❖ Alternative Hypothesis (H_1): There is a significant difference in the mean test scores between at least two teaching methods.

Step 2: Calculate Means

❖ Calculate the mean test score for each group:

$$\text{Mean A} = (85 + 88 + 90 + 92 + 87) / 5 = 88.4$$

$$\text{Mean B} = (78 + 80 + 82 + 85 + 81) / 5 = 81.2$$

$$\text{Mean C} = (88 + 85 + 84 + 87 + 90) / 5 = 86.8$$

Step 3: Calculate Sum of Squares (SSW and SSB) within the Groups / Between the Groups

❖ Let's calculate the sum of squares (SSW and SSB) for the productivity scores in each

$$SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2$$

$$SSB = \sum_{i=1}^k n_i (\bar{X}_i - \bar{X})^2$$

Where:

- X_{ij} is the score of the j-th observation in the i-th group.
- $\bar{X}_i$ is the mean of the i-th group.
- $\bar{X}$ is the overall mean.
- n_i is the number of observations in the i-th group.
- k is the number of groups.

❖ Let's substitute the values into the formulas for SSW_A , SSB_A , SSW_B , SSB_B , SSW_C , and SSB_C to show the calculations

$$SSW_A = \sum_{j=1}^5 (X_{Aj} - \bar{X}_A)^2$$

Where:

- X_{Aj} are the individual scores.
- $\bar{X}_A = 88.4$ is the mean of the scores.

$$SSW_A = (85 - 88.4)^2 + (88 - 88.4)^2 + (90 - 88.4)^2 + (92 - 88.4)^2 + (87 - 88.4)^2$$

$$SSW_A = (-3.4)^2 + (-0.4)^2 + (1.6)^2 + (3.6)^2 + (-1.4)^2$$

$$SSW_A = 11.56 + 0.16 + 2.56 + 12.96 + 1.96$$

$$SSW_A = 29.2$$

- Calculate overall mean ($\bar{X}$):

$$\bar{X} = \frac{85 + 88 + 90 + 92 + 87 + 78 + 80 + 82 + 85 + 81 + 88 + 85 + 84 + 87 + 90}{15} = \frac{1282}{15} = 85.47$$

$$SSB_A = 5 \times (88.4 - 85.47)^2 = 5 \times (2.93)^2 = 5 \times 8.5849 = 42.9245 \approx 42.92$$

For Program B:

$$SSW_B = \sum_{j=1}^5 (X_{Bj} - 81.2)^2$$

$$SSW_B = (78 - 81.2)^2 + (80 - 81.2)^2 + (82 - 81.2)^2 + (85 - 81.2)^2 + (81 - 81.2)^2$$

$$SSW_B \approx (-3.2)^2 + (-1.2)^2 + 0.8^2 + 3.8^2 + (-0.2)^2$$

$$SSW_B \approx 10.24 + 1.44 + 0.64 + 14.44 + 0.04$$

$$SSW_B \approx 26.8$$



$$SSB_B = 5 \times (81.2 - 85.47)^2 = 5 \times (-4.27)^2 = 5 \times 18.2329 = 91.1645 \approx 91.16$$

For Program C:

$$SSW_C = (88 - 86.8)^2 + (85 - 86.8)^2 + (84 - 86.8)^2 + (87 - 86.8)^2 + (90 - 86.8)^2$$

$$SSW_C = (1.2)^2 + (-1.8)^2 + (-2.8)^2 + (0.2)^2 + (3.2)^2$$

$$SSW_C = 1.44 + 3.24 + 7.84 + 0.04 + 10.24 = 22.8$$

$$SS_{BC} = 5 \times (86.8 - 85.47)^2 = 5 \times 1.7689 = 8.84.$$

$$SSW = SSW_A + SSW_B + SSW_C \quad SSW = 29.2 + 26.8 + 22.8 = 78.8$$

$$SSB = SSB_A + SSB_B + SSB_C : \quad SSB = 42.92 + 91.16 + 8.84 = 142.92$$

Step 4: Calculate Degrees of Freedom (DFW and DFB)

$$DFW = N - k = 15 - 3 = 12$$

$$DFB = k - 1 = 3 - 1 = 2$$

Step 5: Calculate Mean Squares (MSW and MSB)

$$MSW = \frac{SSW}{DFW}$$

$$MSW = \frac{78.8}{12} = 6.5667 \approx 6.57$$

$$MSB = \frac{SSB}{DFB}$$

$$MSB = \frac{142.92}{2} = 71.46$$

Step 6: Calculate F Ratio

$$F = \frac{MSB}{MSW} = \frac{71.46}{6.57} \approx 10.88$$

Step 7: Decision Rule

- With $DFB = 2$ and $DFW = 12$ at a significance level of 0.05, the critical F-value is approximately 3.89.
- Since $F > 3.89$ ($10.88 > 3.89$), we reject the null hypothesis.

Step 8: Conclusion

- ❖ There is sufficient evidence to suggest that there are significant differences in the mean test scores between at least two teaching methods.

Important Note:

- ❖ The critical F-value of 3.89 is a hypothetical value used for illustration purposes. F-distribution table or using statistical software

Alternative: Use Python

You can also calculate the critical F-value using Python with the scipy.stats library:

```
from scipy.stats import f
```

```
# Parameters
```

```
alpha = 0.05
```

```
dfb = 2 # Degrees of freedom between groups
```

```
dfw = 12 # Degrees of freedom within groups
```

```
# Critical F-value
```

```
critical_f = f.ppf(1 - alpha, dfb, dfw)
```

```
print("Critical F-Value:", critical_f)
```

```
: from scipy.stats import f

# Parameters
alpha = 0.05
dfb = 2 # Degrees of freedom between groups
dfw = 12 # Degrees of freedom within groups

# Critical F-value
critical_f = f.ppf(1 - alpha, dfb, dfw)
print("Critical F-Value:", critical_f)
```

Critical F-Value: 3.8852938346523933

Performs a One-Way ANOVA (Analysis of Variance) test to determine if there is a significant difference in test scores among three different teaching methods (Method A, Method B, and Method C). The goal is to perform a One-Way ANOVA to test if there is a significant difference in the mean test scores between the three methods. Data:

Method A: 85, 88, 90, 92, 87

Method B: 78, 80, 82, 85, 81

Method C: 88, 85, 84, 87, 90

Python Program

Import Necessary Libraries

1. `import numpy as np`
2. `import scipy.stats as stats`

❖ `scipy.stats (stats)`: Contains statistical functions, including One-Way ANOVA.

Define Test Scores for Each Teaching Method

1. `method_A = [85, 88, 90, 92, 87]`
2. `method_B = [78, 80, 82, 85, 81]`
3. `method_C = [88, 85, 84, 87, 90]`

Perform One-Way ANOVA Test

1. `f_stat, p_value = stats.f_oneway(method_A, method_B, method_C)`

❖ `stats.f_oneway()` performs a One-Way ANOVA test.

❖ It returns:

❖ **F-Statistic (f_stat):** Measures the ratio of variance between groups to variance within groups.

❖ **P-Value (p_value):** Determines if the observed differences are statistically significant.

What is One-Way ANOVA?

- ❖ Compares the means of two or more independent groups in order to determine whether there is statistical evidence that the associated population means are significantly different
- ❖ It is used to compare the means of more than two groups.
- ❖ Null Hypothesis (H_0): The means of all groups are equal (no significant difference).
- ❖ Alternative Hypothesis (H_1): At least one group's mean is different.

Output the Results

1. `print("F-Statistic:", f_stat)`
2. `print("P-Value:", p_value)`

Output:

F-Statistic: 10.88324873096448

P-Value: 0.0020145191028122247

Interpretation of Results

1. `alpha = 0.05` # Significance level
2. `if p_value < alpha:`
3. `print("Reject the null hypothesis: There is a significant difference in the mean test scores between the teaching methods.")`
4. `else:`
5. `print("Fail to reject the null hypothesis: No significant difference in the mean test scores between the teaching methods.")`

Output:

Reject the null hypothesis: There is a significant difference in the mean test scores between the teaching methods.

Two-Way ANOVA

- ❖ Two-Way ANOVA (Analysis of Variance) is a statistical method used to analyze the effects of two independent variables (factors) on a dependent variable. It helps determine whether these factors influence the dependent variable individually (main effects) and whether they interact with each other (interaction effect).

When to Use Two-Way ANOVA?

- ❖ Use the Two-Way ANOVA when:
 - ❖ You have **one continuous dependent variable** (e.g., crop yield, test scores, etc.).
 - ❖ You have **two categorical independent variables** (factors) (e.g., fertilizer type and irrigation level).
 - ❖ You want to **test both the individual effects of each factor and their combined effect**.

Note:

- ❖ The only difference between one-way and two-way ANOVA is the number of independent variables. A one-way ANOVA has one independent variable, while a two-way ANOVA has two.

Components of Two-Way ANOVA

- ❖ Two-Way ANOVA tests for three effects:
 - ❖ **Main Effect of Factor A** → Does Factor A significantly impact the dependent variable?
 - ❖ **Main Effect of Factor B** → Does Factor B significantly impact the dependent variable?
 - ❖ **Interaction Effect ($A \times B$)** → Do Factor A and Factor B interact to affect the dependent variable differently?

How Two-Way ANOVA Works?

- ❖ Two-Way ANOVA tests whether two independent categorical variables (factors) affect a dependent variable significantly and whether they interact with each other.
- ❖ It does this by comparing the variance within groups to the variance between groups.

Steps in Two-Way ANOVA Testing

1. Formulate Hypotheses

- ❖ **Main Effect of Factor A:** Does Factor A (e.g., fertilizer type) influence the dependent variable?
- ❖ **Main Effect of Factor B:** Does Factor B (e.g., irrigation level) influence the dependent variable?
- ❖ **Interaction Effect ($A \times B$):** Does Factor A's effect depend on Factor B?

2. Collect and Organize Data

- ❖ The dependent variable must be continuous (e.g., plant height, yield).
- ❖ The two independent variables must be categorical (e.g., fertilizer type: Organic/Chemical, irrigation level: Low/Medium/High).

3. Calculate Group Means and Overall Mean

- ❖ Compute the mean of the dependent variable for each level of Factor A and Factor B.
- ❖ Compute the overall mean (grand mean).

4. Calculate Variances (Sum of Squares - SS)

- ❖ **SS(A)**: Variation due to Factor A (e.g., fertilizer type).
- ❖ **SS(B)**: Variation due to Factor B (e.g., irrigation level).
- ❖ **SS(Interaction)**: Variation due to the interaction of A and B.
- ❖ **SS(Error)**: Variation within groups (random error).

5. Compute F-Statistics

$$F = \frac{\text{Mean Square of Factor}}{\text{Mean Square of Error}}$$

- ❖ A higher F-value indicates that the factor significantly affects the dependent variable.

6. Compare with Critical Value (p-value)

- ❖ If $p < 0.05$, reject H_0 , meaning the factor has a significant effect.
- ❖ If $p > 0.05$, fail to reject H_0 , meaning no significant effect.

7. Interpret the Results

- ❖ If both main effects are significant $\rightarrow$ Both factors independently affect the dependent variable.
- ❖ If the interaction effect is significant $\rightarrow$ The effect of one factor depends on the level of the other factor.

Scenario: Effect of Fertilizer and Irrigation on Papaya Plant Growth using Two-Way ANOVA

Objective

- ❖ We want to analyze how fertilizer type and irrigation level affect papaya plant height using Two-Way ANOVA. Additionally, we check if these two factors interact to influence plant growth.

Aim of the Program

- ❖ The aim of this program is to analyze the impact of fertilizer type (Organic vs. Chemical) and irrigation level (Low, Medium, High) on papaya plant growth using Two-Way ANOVA. The program also tests whether there is an interaction effect between fertilizer type and irrigation level that influences plant height.

Step 1: Define the Variables

- ❖ Dependent Variable: Papaya plant height (in cm)
- ❖ Factor A (Fertilizer Type):
 - ❖ Organic
 - ❖ Chemical
- ❖ Factor B (Irrigation Level):
 - ❖ Low
 - ❖ Medium
 - ❖ High

Step 2: Collect Data

- ❖ We conduct an experiment where we measure the height of papaya plants under different fertilizer and irrigation conditions.

Fertilizer Type	Low Irrigation	Medium Irrigation	High Irrigation
Organic	45 cm, 47 cm, 44 cm	50 cm, 52 cm, 49 cm	55 cm, 57 cm, 56 cm
Chemical	50 cm, 51 cm, 49 cm	55 cm, 56 cm, 54 cm	60 cm, 61 cm, 59 cm

Step 3: Hypotheses

Main Effects:

Fertilizer Effect (Factor A):

H_0 : Fertilizer type has no effect on papaya plant height.

H_1 : Fertilizer type affects papaya plant height.

Irrigation Effect (Factor B):

H_0 : Irrigation level has no effect on papaya plant height.

H_1 : Irrigation level affects papaya plant height.

Interaction Effect:

- ❖ H_0 : No interaction between fertilizer and irrigation (one factor doesn't depend on the other).
- ❖ H_1 : There is an interaction between fertilizer and irrigation (one factor influences the effect of the other).

Step 4: Perform Two-Way ANOVA

1. Import Required Libraries

1. `import pandas as pd`
2. `import statsmodels.api as sm`
3. `import statsmodels.formula.api as smf`
4. `from scipy import stats`

- ❖ **pandas**: Used for handling structured data (tables).
- ❖ **statsmodels.api**: Provides statistical models and tests.
- ❖ **statsmodels.formula.api**: Allows for defining models using formulas.
- ❖ **scipy.stats**: Provides statistical functions.

2. Create Sample Data

```
1. data = {  
2.     "Fertilizer": ["Organic", "Organic", "Organic", "Organic", "Organic", "Organic",  
3.         "Chemical", "Chemical", "Chemical", "Chemical", "Chemical", "Chemical"],  
4.     "Irrigation": ["Low", "Medium", "High", "Low", "Medium", "High",  
5.         "Low", "Medium", "High", "Low", "Medium", "High"],  
6.     "Height": [50, 55, 60, 52, 58, 63, 45, 50, 55, 48, 53, 57] # Example heights in cm  
7. }
```

- ❖ The dataset contains **12 observations**.
- ❖ Two categorical variables:
 - ❖ **Fertilizer** (Organic vs. Chemical)
 - ❖ **Irrigation** (Low, Medium, High)
- ❖ One numerical variable:
 - ❖ **Height** (measured in cm)

3. Convert Data to a DataFrame

1. `df = pd.DataFrame(data)`

❖ Converts the dictionary data into a **pandas DataFrame** for easier analysis

4. Define the ANOVA Model

1. `model = smf.ols('Height ~ C(Fertilizer) + C(Irrigation) + C(Fertilizer):C(Irrigation)', data=df).fit()`

❖ `ols()` stands for Ordinary Least Squares regression, a method used to estimate relationships between variables.

❖ In ANOVA, OLS regression is used to compute group means and compare them statistically.

- ❖ `ols('Height ~ C(Fertilizer) + C(Irrigation) + C(Fertilizer):C(Irrigation)', data=df).fit():`
- ❖ **C(Fertilizer)** → Effect of **fertilizer** on height.
- ❖ **C(Irrigation)** → Effect of **irrigation** on height.
- ❖ **C(Fertilizer):C(Irrigation)** → Interaction effect between fertilizer and irrigation.

5. Perform Two-Way ANOVA

1. `anova_table = sm.stats.anova_lm(model, typ=2)`

- ❖ `sm.stats.anova_lm(model, typ=2)`: Performs **Two-Way ANOVA**
- ❖ **Type 2 ANOVA** is used, which considers each effect separately after accounting for other effects.

6. Display ANOVA Table

1. `print("ANOVA Results:")`
2. `print(anova_table)`

ANOVA Results:

	sum_sq	df	F	PR(>F)
C(Fertilizer)	75.000000	1.0	20.454545	0.004006
C(Irrigation)	200.166667	2.0	27.295455	0.000971
C(Fertilizer):C(Irrigation)	0.500000	2.0	0.068182	0.934804
Residual	22.000000	6.0	NaN	NaN

❖ The ANOVA table contains:

- ❖ **Sum of Squares (SS):** Variability explained by each factor.
- ❖ **Degrees of Freedom (df):** Number of independent values.
- ❖ **F-statistic:** Measures variance explained relative to residual variance.
- ❖ **p-value (PR(>F)):** Indicates statistical significance.

7. Check for Statistical Significance

1. `significance_level = 0.05`
2. `print("\nStatistical Significance:")`
3. `for factor in anova_table.index:`
4. `p_value = anova_table.loc[factor, 'PR(>F)']`
5. `if p_value < significance_level:`
6. `print(f"{factor}: Significant effect (p-value = {p_value:.4f})")`
7. `else:`
8. `print(f"{factor}: Not significant (p-value = {p_value:.4f})")`

`Statistical Significance:`

`C(Fertilizer): Significant effect (p-value = 0.0040)`

`C(Irrigation): Significant effect (p-value = 0.0010)`

`C(Fertilizer):C(Irrigation): Not significant (p-value = 0.9348)`

`Residual: Not significant (p-value = nan)`

Compares p-values with a significance level (0.05).

- ❖ If $p\text{-value} < 0.05 \rightarrow$ Factor has a significant effect.
- ❖ If $p\text{-value} \geq 0.05 \rightarrow$ Factor is not statistically significant.

Conclusion:

- ❖ The Two-Way ANOVA analysis shows that both fertilizer type and irrigation level have significant effects on papaya plant height, indicating that changing either factor influences plant growth. **However, the interaction between fertilizer and irrigation is not statistically significant**, meaning their combined effect does not create an additional impact beyond their individual effects. This suggests that while both factors are important, they work independently rather than synergistically to affect plant height.

POST HOC TEST ANOVA

- ❖ A post hoc test, also known as a post hoc analysis or pairwise comparison, is conducted after an analysis of variance (ANOVA) to determine specific group differences when the overall ANOVA results indicate a significant difference among three or more groups.
- ❖ ANOVA is a statistical method used to analyze the variation between group means in a dataset to determine if there are statistically significant differences among the groups.
- ❖ When the ANOVA test indicates a significant difference, it doesn't specify which groups are different from each other.
- ❖ Post hoc tests are employed to make pairwise comparisons between individual groups and identify where the differences lie.
- ❖ The need for post hoc tests arises because conducting multiple pairwise comparisons without adjustment increases the likelihood of making a Type I error (incorrectly rejecting a true null hypothesis).

When to Use a Post Hoc Test?

- ❖ If ANOVA results are statistically significant (i.e., $p\text{-value} < 0.05$).
- ❖ When multiple groups (more than two) are being compared.
- ❖ To control for Type I errors (false positives) that arise from multiple comparisons.

Application of Post Hoc Tests in ANOVA

- ❖ Post hoc tests are widely used in various fields where multiple groups need to be compared. Here are some real-world applications
- ❖ **Medical Research** – Comparing the effectiveness of different prostate cancer treatments.
- ❖ **Agriculture** – Analyzing the impact of different fertilizers on papaya growth.
- ❖ **Business & Marketing** – Evaluating customer satisfaction across product versions.
- ❖ **Education** – Identifying the best teaching method based on student performance.
- ❖ **Sports Science** – Comparing the effectiveness of various training programs on athletes

❖ Scenario with Example Data

- ❖ A farmer tests three fertilizers (A, B, and C) on papaya plants and measures their height after three months. ANOVA reveals a significant difference in growth among the groups. A post hoc test (Tukey's HSD) identifies which fertilizer performs best.

Fertilizer	Plant Height (cm)
A	45
A	50
A	48
B	60
B	62
B	65
C	55
C	53
C	52

- ❖ Step-by-Step Explanation of ANOVA and Tukey's HSD Test in PythonWe are analyzing the effect of three fertilizers (A, B, and C) on plant growth using ANOVA and Tukey's HSD post hoc test.

Step 1: Import Necessary Libraries

1. `import pandas as pd`
2. `import scipy.stats as stats`
3. `import statsmodels.api as sm`
4. `from statsmodels.formula.api import ols`
5. `from statsmodels.stats.multicomp import pairwise_tukeyhsd`

Step 2: Create Example Data

- ❖ We have three groups (A, B, and C) with plant height values after applying different fertilizers.

Example dataset

1. data = {
2. 'Fertilizer': ['A', 'A', 'A', 'B', 'B', 'B', 'C', 'C', 'C'],
3. 'Height': [45, 50, 48, 60, 62, 65, 55, 53, 52]
4. }

Create a DataFrame

1. df = pd.DataFrame(data)

❖ **Step 3: Perform One-Way ANOVA checks if there is at least one significant difference between the groups.**

Fit the ANOVA model

1. `model = ols('Height ~ C(Fertilizer)', data=df).fit()`
2. `anova_table = sm.stats.anova_lm(model, typ=2)`

`ols('Height ~ C(Fertilizer)', data=df).fit()` → Creates a linear model for ANOVA.

Print ANOVA result

1. `print("### ANOVA Result ###")`
2. `print(anova_table)`

```
### ANOVA Result ###
              sum_sq    df      F    PR(>F)
C(Fertilizer)  328.222222  2.0  32.822222  0.000587
Residual       30.000000  6.0      NaN      NaN
```

Step 4: Check p-value for Significance

❖ If $p < 0.05$, we proceed to the Tukey's HSD test.

Extract p-value from ANOVA table

1. `p_value = anova_table["PR(>F)"][0]`
2. `alpha = 0.05 # Significance level`
3. `if p_value < alpha:`
4. `print("\nStatistically Significant Difference Found (p < 0.05)")`

Step 5: Perform Tukey's HSD Test

1. `tukey = pairwise_tukeyhsd(df['Height'], df['Fertilizer'], alpha=0.05)`
2. `print("\n### Tukey HSD Test Result ###")`
3. `print(tukey)`
4. `else:`
5. `print("\nNo Significant Difference Found (p >= 0.05)")`

Statistically Significant Difference Found ($p < 0.05$)

Tukey HSD Test Result

Multiple Comparison of Means - Tukey HSD, FWER=0.05

```
=====
group1 group2 meandiff p-adj    lower    upper  reject
-----
      A      B  14.6667 0.0005    9.0648  20.2685   True
      A      C   5.6667 0.0479    0.0648  11.2685   True
      B      C   -9.0 0.0063  -14.6019  -3.3981   True
-----
```

Step 5: Perform Tukey's HSD Post Hoc Test

❖ Tukey's HSD test tells us which specific groups are different.

1. `tukey = pairwise_tukeyhsd(df['Height'], df['Fertilizer'], alpha=0.05)`
2. `print("\n### Tukey HSD Test Result ###")`
3. `print(tukey)`

❖ The function `pairwise_tukeyhsd` is used to perform Tukey's Honestly Significant Difference (HSD) test, which is a post hoc test used after ANOVA to determine which specific groups are significantly different from each other

```
### Tukey HSD Test Result ###
Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1 group2 meandiff p-adj  lower  upper  reject
-----
A      B      14.6667 0.0005   9.0648 20.2685   True
A      C       5.6667 0.0479   0.0648 11.2685   True
B      C      -9.0 0.0063 -14.6019 -3.3981   True
-----
```

Step 6: Final Conclusion

- ❖ ANOVA Result $\rightarrow p = 0.0027 (< 0.05)$, so there is a significant difference between fertilizers.
- ❖ Tukey's HSD Results
- ❖ A vs B: $p = 0.002 \rightarrow$ Significant difference (B performs better).
- ❖ A vs C: $p = 0.136 \rightarrow$ No significant difference.
- ❖ B vs C: $p = 0.015 \rightarrow$ Significant difference (B performs better).

Final Conclusion Significant:

- ❖ Fertilizer B is the most effective, as it significantly increases plant height compared to A and C.

Aim of the Program:

- ❖ To analyze the effect of different fertilizers on plant growth using ANOVA and Tukey's HSD test to identify significant differences between fertilizer groups

Conclusion of the Program:

- ❖ The ANOVA test indicates a significant difference between fertilizers ($p < 0.05$). Tukey's HSD test confirms that Fertilizer B significantly increases plant height compared to A and C, while there is no significant difference between A and C.

Artificial Intelligence

- ❖ The Father of Artificial Intelligence is **John McCarthy**. He was an American computer scientist who coined the term "Artificial Intelligence" (AI) in **1956** and played a crucial role in its development.
- ❖ John McCarthy developed the LISP (LISt Processing) programming language, which became essential for AI research. He also contributed significantly to machine learning, automated reasoning, and early AI systems.
- ❖ **Artificial Intelligence (AI)** is the simulation of human intelligence in machines to perform tasks like learning, reasoning, and problem-solving
- ❖ **Machine Learning (ML)** is a subset of AI that enables systems to learn from data and improve performance without explicit programming - Linear Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), K-Nearest Neighbors (KNN)
- ❖ **Deep Learning (DL)** is a subset of machine learning that uses multi-layered neural networks to automatically learn patterns from large amounts of data. - Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs) , Neural Turing Machines (NTMs)

Revolutions of AI

1st AI Revolution (1950s – 1970s) – Birth of AI

- ❖ Introduction of AI as a field (John McCarthy coined the term "AI" in 1956).
- ❖ Development of early AI programs and rule-based systems.
- ❖ Creation of LISP, the first AI programming language.
- ❖ **Limitations:** Lack of computing power and real-world applications.

2nd AI Revolution (1980s – 2000s) – Machine Learning & Expert Systems

- ❖ Rise of expert systems, which used if-then rules to make decisions.
- ❖ Development of neural networks and machine learning (ML) techniques.
- ❖ Introduction of algorithms like decision trees and support vector machines (SVMs).
- ❖ **Limitations:** Required a lot of labeled data and had limited generalization.

3rd AI Revolution (2010s – Present) – Deep Learning & Big Data

- ❖ Breakthroughs in deep learning with neural networks like CNNs & RNNs.
- ❖ Use of big data and cloud computing to train AI models efficiently.
- ❖ Rise of AI applications like self-driving cars, voice assistants (Siri, Alexa), and recommendation systems.
- ❖ AI now powers industries such as healthcare, finance, agriculture, and robotics.
- ❖ **Limitations:** High data dependency, lack of explain ability, computational costs, bias, security risks, ethical concerns, and limited generalization beyond training data.

4th AI Revolution (Future) – Artificial General Intelligence (AGI) & Beyond

- ❖ Development of AGI – AI that can think, learn, and reason like humans.
- ❖ AI will become self-aware and autonomous in decision-making.
- ❖ Integration of AI with quantum computing for faster processing
- ❖ Ethical and regulatory challenges will increase as AI takes on more human-like roles.
- ❖ **Limitations:** Ethical dilemmas, loss of human control, security risks, unpredictability, existential threats, and challenges in aligning AI with human values

Broad Operations in AI, ML, and DL

Operation	Machine Learning (ML)	Deep Learning (DL)
Regression	Linear Regression, Decision Trees	Deep Neural Networks (DNNs) for regression
Classification	Logistic Regression, SVM, Random Forest	CNNs for image classification, Transformers for NLP
Clustering	K-Means, Hierarchical Clustering	Autoencoders, Self-Organizing Maps (SOMs)
Dimensionality Reduction	PCA, t-SNE, LDA	Autoencoders, Variational Autoencoders (VAEs)
Reinforcement Learning	Q-Learning, SARSA	Deep Q-Networks (DQN), Proximal Policy Optimization (PPO)

Broad Operations in AI, ML, and DL

Operation	Machine Learning (ML)	Deep Learning (DL)
Anomaly Detection	Isolation Forest, One-Class SVM	LSTMs for anomaly detection, Autoencoders
Generative Modeling	Gaussian Mixture Models (GMMs)	GANs, Variational Autoencoders (VAEs)
Feature Selection & Extraction	Recursive Feature Elimination (RFE), Mutual Information	CNN feature extraction, Transfer Learning
Optimization & Hyperparameter Tuning	Grid Search, Random Search	Bayesian Optimization, Genetic Algorithms
Time Series Forecasting	ARIMA, Exponential Smoothing	LSTMs, Transformer-based models

Broad Operations in AI, ML, and DL

Operation	Machine Learning (ML)	Deep Learning (DL)
Sequence Prediction	Markov Models, Hidden Markov Models (HMMs)	RNNs, LSTMs, GRUs
Recommender Systems	Collaborative Filtering, Content-Based Filtering	Deep Learning-based recommendation engines (e.g., Neural Collaborative Filtering)
Object Detection & Recognition	HOG + SVM, Viola-Jones Algorithm	YOLO, Faster R-CNN, SSD
Speech & Audio Processing	MFCC + HMM, GMM	WaveNet, DeepSpeech, Transformer-based models
Graph-Based Learning	PageRank, Community Detection	Graph Neural Networks (GNNs), Graph Convolutional Networks (GCNs)
Neural Architecture Search (NAS)	Grid Search, Random Search	AutoML, Reinforcement Learning-based NAS

Regression: Simple Linear Regression

- ❖ Machine learning is a subset of AI, which enables the machine to automatically learn from data, improve performance from past experiences, and make predictions.
- ❖ Machine learning contains a set of algorithms that work on a huge amount of data. Data is fed to these algorithms to train them, and on the basis of training, they build the model & perform a specific task.
- ❖ Machine learning (ML) is a type of artificial intelligence (AI) focused on building computer systems that learn from data.
- ❖ The broad range of techniques ML encompasses enables software applications to improve their performance over time.
- ❖ Machine learning algorithms are trained to find relationships and patterns in data.
- ❖ They use historical data as input to **make predictions, classify information, cluster data points**, reduce dimensionality and even help generate new content.

Regression: Simple Linear Regression

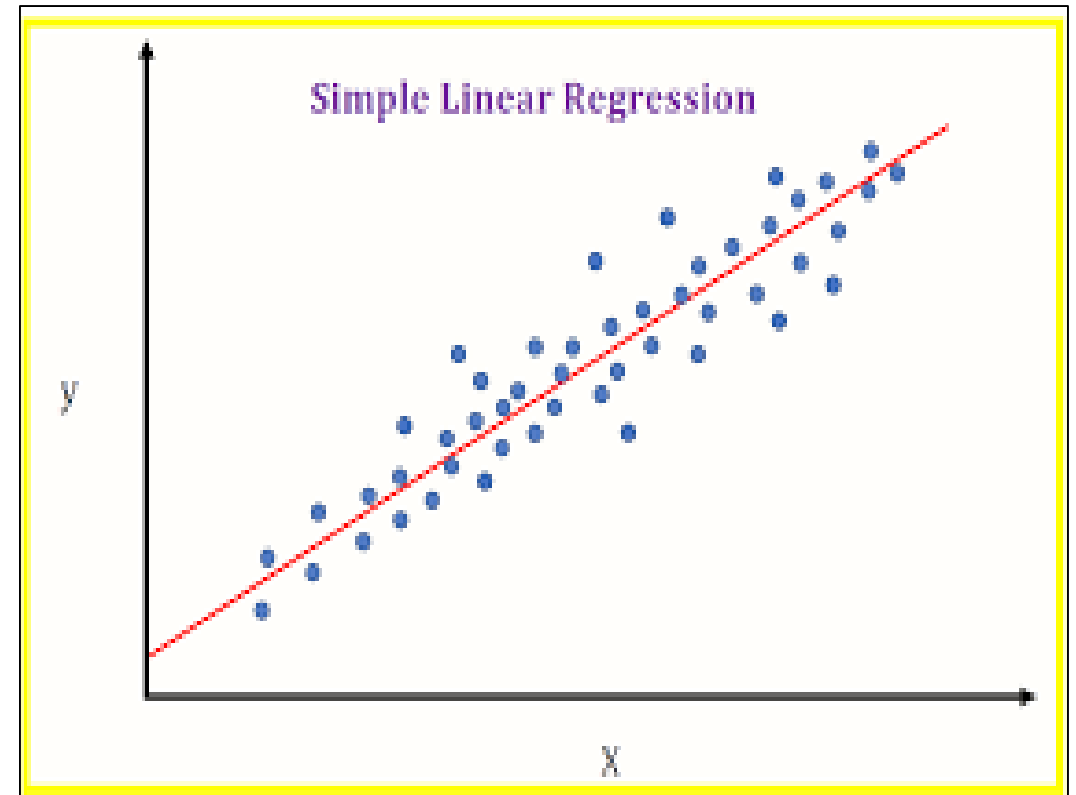
- ❖ Linear regression predicts the relationship between two variables by assuming a linear connection between the independent and dependent variables.
- ❖ It try to find the optimal line that minimizes the sum of squared differences between **predicted and actual values**.
- ❖ Applied in various domains like economics and finance, this method analyzes and forecasts data trends. It can extend to multiple linear regression involving several independent variables and logistic regression, suitable for binary classification problems
- ❖ **Linear regression is a Machine Learning model that depends on the linear relationship between a dependent variable and one or more independent variables.**

Problem Statement example for Simple Linear Regression:

- ❖ Here we are taking a dataset that has two variables: salary (dependent variable) and experience (Independent variable). The goals of this problem is:
- ❖ We want to find out if there is any correlation between these two variables
- ❖ We will find the best fit line for the dataset.
- ❖ How the dependent variable is changing by changing the independent variable.

- ❖ Let us understand the word ‘linear relationship’. We can say two variables are in linear relationship when their values can be represented using a straight line. That means, the data points (or values of the variables) lie on the straight line.
- ❖ **When there is only one independent variable, it is called Simple Linear regression.**
- ❖ **For more than one independent variables, the model is called Multiple Linear regression.**

Simple linear regression is a regression model that estimates the relationship between one independent variable and one dependent variable using a straight line. Both variables should be quantitative.



Finding the best fit line:

- ❖ When working with linear regression, our main goal is to **find the best fit line** that means the error between predicted values and actual values should be minimized. **The best fit line will have the least error.**
- ❖ The different values for weights or the coefficient of lines (a_0 , a_1) gives a different line of regression, so we need to calculate the best values for a_0 and a_1 to find the best fit line, so to calculate this we use cost function.

Cost function:

- ❖ The different values for weights or coefficient of lines (a_0 , a_1) gives the different line of regression, and the cost function is used to estimate the values of the coefficient for the best fit line.
- ❖ Cost function optimizes the regression coefficients or weights. It measures how a linear regression model is performing.
- ❖ We can use the cost function to find the accuracy of the mapping function, which maps the input variable to the output variable. This mapping function is also known as Hypothesis function.

- ❖ For Linear Regression, we use the **Mean Squared Error (MSE) cost function**, which is the average of squared error occurred between the predicted values and actual values. It can be written as:
- ❖ For the above linear equation, MSE can be calculated as:

$$MSE = \frac{1}{N} \sum_{i=1}^n (y_i - (a_1 x_i + a_0))^2$$

Where,

N=Total number of observation, y_i = Actual value, $(a_1 x_i + a_0)$ = Predicted value.

- ❖ **Residuals: Residuals help you understand how well a regression model fits a data set.**
- ❖ **The distance between the actual value and predicted values is called residual.**
- ❖ **If the observed points are far from the regression line, then the residual will be high, and so cost function will high.**
- ❖ **If the scatter points are close to the regression line, then the residual will be small and hence the cost function.**

R-squared method:

- ❖ R-squared is a statistical method that determines the goodness of fit.
- ❖ It measures the strength of the relationship between the dependent and independent variables on a scale of 0-100%.
- ❖ The **high value of R-square** determines the less difference between the predicted values and actual values and hence **represents a good model**.
- ❖ It is also called a coefficient of determination, or coefficient of multiple determination for multiple regression.
- ❖ It can be calculated from the below formula:

$$\text{R-squared} = \frac{\text{Explained variation}}{\text{Total Variation}}$$

$$\begin{aligned} R^2 &= 1 - \frac{\text{Residual variance}}{\text{Total variance}} \\ &= \frac{\text{Total variance} - \text{Residual variance}}{\text{Total variance}} \\ &= \frac{\text{Explained variance}}{\text{Total variance}} \\ &= \text{Fraction of total variance explained} \end{aligned}$$

Gradient Descent:

- ❖ Gradient descent is used to minimize the MSE by calculating the gradient of the cost function.
- ❖ A regression model uses gradient descent to update the coefficients of the line by reducing the cost function.
- ❖ **It is done by a random selection of values of coefficient and then iteratively update the values to reach the minimum cost function.**
- ❖ Gradient descent is an optimization algorithm which is commonly-used to train machine learning models and neural networks. It trains machine learning models by minimizing errors between predicted and actual results.

The Linear Equation

❖ In mathematics, to draw a line, we use the equation:

$$y = mx + b$$

❖ Where **m** is slope and **b** is a constant.

❖ **This equation is useful to find y value depending on x value.** Here, **y** is called the **dependent variable** and **x** is known as **independent variable**.

❖ In Statistics, we write the same equation as:

❖ Here, the **slope is β_1** . The **constant value β_0** is called intercept. β_0 , indicates the distance on y axis.

❖ Let us take a linear equation: **$y = 4 + 2x$** . Compare this with **$Y = \beta_0 + \beta_1 X$** .

❖ Here, **$\beta_0 = 4$** , **axis** and **$\beta_1 = 2$** . By substituting x value into this equation, we can find the value of y.

❖ So, x is called independent variable and y is called dependent variable since **y value is dependent on x value**.

$$y = 4 + 2x$$

If $x = 0$, then $y = 4 + 2(0) = 4$, If $x = 2$, then $y = 4 + 2(2) = 8$.

If $x = 4$, then $y = 4 + 2(4) = 12$, If $x = 6$, then $y = 4 + 2(6) = 16$.

- ❖ In this manner, when x is increased in steps of 2, y values are increasing in steps of 4. These values are shown in the **Figure 1**.

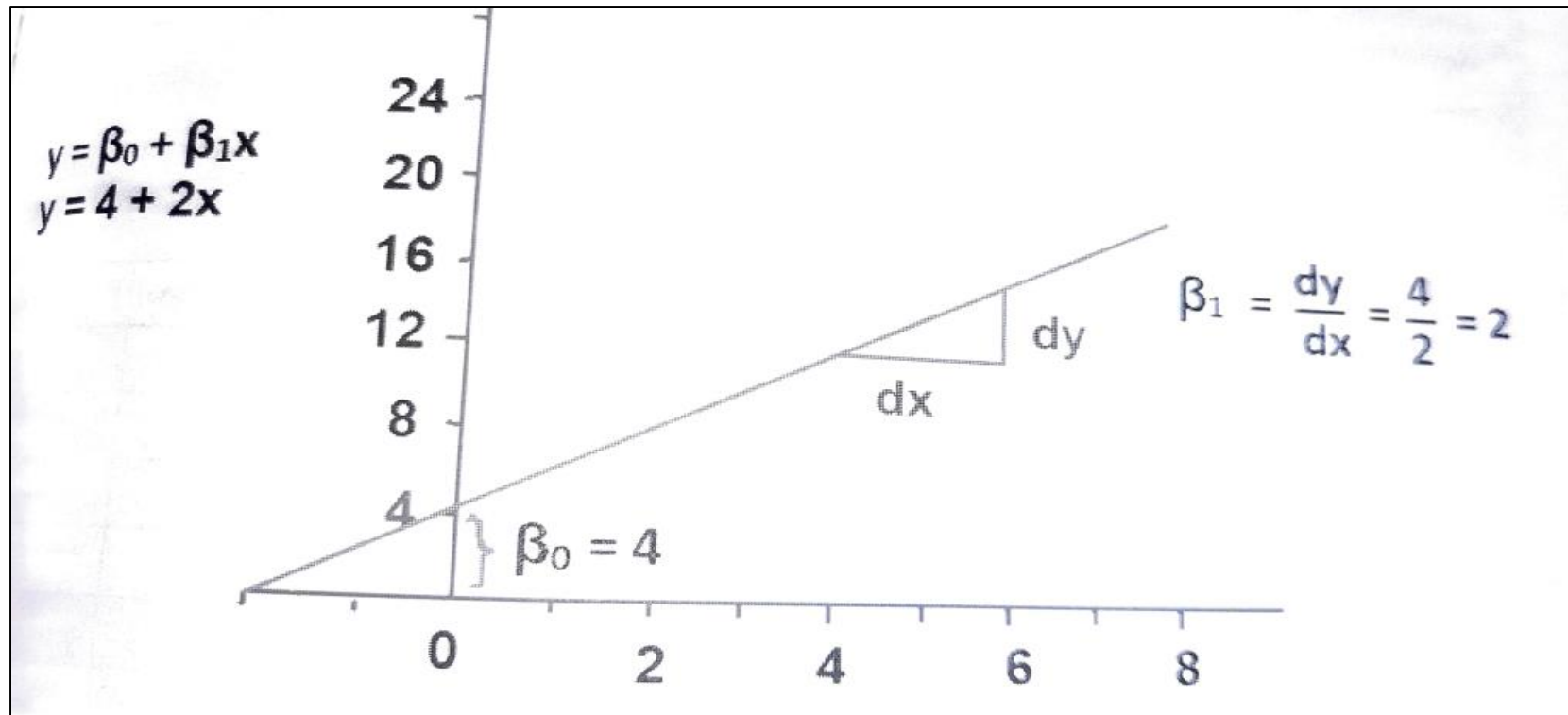


Figure 1: Understanding Linear Equation

- ❖ From the Figure, we can calculate the **slope β_1** , and **intercept β_0** , values, as:
- ❖ **Slope $\beta_1 = \text{deviation in } y / \text{deviation in } x = dy / dx = 4 / 2 = 2$**
- ❖ **Intercept $\beta_0 = \text{distance on } y \text{ axis where the line crosses} = 4$.**
- ❖ **Why we called $y = 4 + 2x$ a linear equation?** Because when the x values and y values are shown in the form of a graph as in Figure 1, it will show a straight line.
- ❖ **That means the relation between independent variable [x] and dependent variable [y] is linear.**
- ❖ **When such a relation exists between the data, we can apply Linear regression analyze the data.**

Squared Value

- ❖ After reading the data from the dataset, we can plot them in the form of a graph. The points may not be exactly on the straight line. There will be deviations from the straight line. This is called error E. The linear regression should consider this error also Hence the formula will be:

$$Y = \beta_0 + \beta_1 X + E$$

- ❖ Let us discuss about this error term "E" now. Suppose the expected values of y are given as:

$$y = [1, 2, 3, 4, 5]$$

- ❖ But the y values are deviated due to the deviations of the data points from the straight line. These deviated y values are:

$$y_1 = [0.8, 2.5, 3, 4.8, 4.4]$$

- ❖ That means we should get 1 but we got 0.8 as y value. This difference is called **residual error (E_1)**.
- ❖ **We should square these errors. If we do not square them, while finding their total, the positive and negative values may cancel out. Hence squaring is needed (E_1^2).**

- ❖ Similarly, we have to calculate the differences of y values from their mean. These are deviations from the mean value ($E_2 = y - \text{mean}$). We have to square this value (E_2^2).
- ❖ **Now, r squared value = 1- (Sum of E_1^2 / Sum of E_2^2)**
- ❖ The above formula can be used to find the value of r squared.
- ❖ Now observe the following table to understand how to calculate r squared value.

y	y_1	$E_1 = y - y_1$	E_1^2	$E_2 = y - \text{Mean}$	E_2^2
1	0.8	0.2	0.04	-2	4
2	2.5	-0.5	0.25	-1	1
3	3	0	0	0	0
4	4.8	-0.8	0.64	1	1
5	4.4	0.6	0.36	2	4
Mean = 3			Sum1 = 1.29		Sum2= 10

❖ Using the above table data, the formula will be:

$$\mathbf{r \ squared = 1 - (Sum1 \ / \ Sum2) = 1 - (1.29 \ / \ 10)=0.871}$$

❖ r squared value is also called ‘Coefficient of determination’.

❖ The r squared value obtained by us is 0.871. In percentage, it is $0.871 \times 100 = 87.1$. This indicates 87% accuracy level for the model.

❖ That means the Linear regression model in this example can explain 87% of deviations for successfully, whereas the remaining 23% cannot be explained by the model. That means there is chance of error level at 23%. The accuracy level of the model is 87%.

❖ **R squared value will be in the range of 0 to 1. If r squared value is closer to 1, then the actual value and predicted values (on the line) will be very close. It represents high accuracy of the model.**

❖ **When r squared value is nearer to 0, they are much apart. So, the prediction may not be correct.**

Note:

The coefficient of determination (R^2) is a number between 0 and 1 that measures how well a statistical model predicts an outcome

- ❖ **Based on the r squared value prediction will change.**
- ❖ The coefficient of determination is a statistical measurement that examines how differences in one variable can be explained by the difference in a second variable, when predicting the outcome of a given event.
- ❖ In other words, this coefficient, which is more commonly known as R-squared (or R^2), assesses how strong the linear relationship is between two variables.
- ❖ What happens when there are no deviations between **y** and **y1**. That means, $E1$ value will be 0. Then $E1^2$ will also be 0.
- ❖ Then the sum of squares of errors (Sum1) will be 0.
- ❖ This indicates 100% accuracy for the model. So, the point is this: in Linear regression models, the sum of squares should have least value and that represents high accuracy.
- ❖ This is the reason, '**Simple Linear Regression model**' is also called Least Squares Regression model'.

- ❖ The following Python code explains how to calculate r squared value for the above example. 'sklearn' is a package from scikit-learn.org that contains many machine learning related modules.
- ❖ In sklearn, we have a module by the name metrics. This module contains a function r2_score(). By calling this function and passing the original data and predicted data, we can find the r squared value as shown below:

❖ **Scikit-learn (Sklearn) is the most useful and robust library for machine learning in Python.**

- ❖ It provides a selection of efficient tools for machine learning and statistical modeling including classification, regression, clustering and dimensionality reduction via a consistent interface in Python.

```
from sklearn.metrics import r2_score
y =[1, 2, 3, 4, 5]
y1 =[0.8,2.5,3,4.8,4.4]
R_Square = r2_score (y, y1)
print('Coefficient of Determination', R_Square)
```

Coefficient of Determination 0.871

Simple Linear Regression for Predicting Salary Based on Experience

Importing Libraries

1. `import pandas as pd`
2. `import numpy as np`
3. `import matplotlib.pyplot as plt`
4. `from sklearn.model_selection import train_test_split`
5. `from sklearn.linear_model import LinearRegression`
6. `from sklearn.metrics import mean_squared_error`

- ❖ pandas is used for data manipulation and data analysis.
- ❖ numpy is used for numerical operations.
- ❖ matplotlib.pyplot is used for data visualization.
- ❖ train_test_split from sklearn.model_selection is used to split the dataset into training and testing sets.
- ❖ LinearRegression from sklearn.linear_model is used to perform linear regression.
- ❖ mean_squared_error from sklearn.metrics is used to calculate Mean Squared Error.

Loading the Dataset:

```
1. data = pd.read_csv("C:\Exp\DA\salary_data.csv")
```

- ❖ This line reads the dataset from the file "Salary_Data.csv" into a pandas DataFrame called data.

Out[5]:

	YearsExperience	Salary
0	1.1	39343
1	1.3	46205
2	1.5	37731
3	2.0	43525

Extracting Features and Target Variable:

```
1. X = data.iloc[:, :-1].values  
2. y = data.iloc[:, 1].values
```

- ❖ X contains the features (independent variables), which are the 'YearsExperience' column from the dataset.
- ❖ y contains the target variable (dependent variable), which is the 'Salary' column from the dataset.
- ❖ data.iloc[:, :-1] selects all rows and all columns except the last one from the DataFrame data.
- ❖ This means we're selecting all the features (independent variables) in the dataset, except for the last column, which is typically the target variable.

Splitting the Dataset:

1. `X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0)`

- ❖ splits the dataset into training and testing sets.
- ❖ `test_size=0.2` indicates that 20% of the data will be used for testing, and the rest for training.
- ❖ `random_state=0` ensures reproducibility by fixing the random seed.

Training the Model:

1. `regressor = LinearRegression()`

2. `regressor.fit(X_train, y_train)`

- ❖ `LinearRegression` is instantiated to create a linear regression model.
- ❖ `fit()` is used to train the model on the training data (`X_train` and `y_train`).

Making Predictions:

1. `y_pred = regressor.predict(X_test)`

❖ `predict()` is used to make predictions on the test data (`X_test`).

Calculating Mean Squared Error (MSE):

1. `mse = mean_squared_error(y_test, y_pred)`

❖ `mean_squared_error()` calculates the mean squared error between the actual target values (`y_test`) and the predicted values (`y_pred`).

Intercept and Coefficient:

1. `intercept = regressor.intercept_`

2. `coefficient = regressor.coef_[0]`

❖ The intercept and coefficient of the linear regression model are obtained from the `intercept_` and `coef_` attributes of the regressor object, respectively.

Predicting a new value

1. `new_experience = 5`
2. `# Example: predicting salary for someone with 5 years of experience`
3. `predicted_salary = regressor.predict([[new_experience]])`
4. `print("Predicted Salary for {} years of experience: {:.2f}".format(new_experience, predicted_salary[0]))`

- ❖ This code predicts the salary for a new value of years of experience (`new_experience`) using the trained linear regression model (`regressor`).
- ❖ Replace `new_experience` with the desired value for prediction. The predicted salary is printed to the console.

Visualizing the Training Set Results:

1. `print("MSE:", mse)`
2. `print("Intercept:", intercept)`
3. `print("Coefficient:", coefficient)`

MSE: 12823412.298126562

Intercept: 26780.099150628186

Coefficient: 9312.575126729187

4. `plt.scatter(X_train, y_train, color='red')`
5. `plt.plot(X_train, regressor.predict(X_train), color='blue')`
6. `plt.title('Salary vs Experience (Training set)')`
7. `plt.xlabel('Years of Experience')`
8. `plt.ylabel('Salary')`
9. `plt.show()`



Simple Linear Regression for Predicting Salary Based on Experience

Aim:

- ❖ To implement Simple Linear Regression using Python to predict an employee's salary based on years of experience by training a model on a given dataset and evaluating its performance.

Conclusion:

- ❖ This program implemented Simple Linear Regression to predict salary based on years of experience. The model was trained, tested, and evaluated using Mean Squared Error (MSE) to assess accuracy. The regression equation enabled salary predictions for new experience values, and visualization confirmed that the model effectively captured the data trend.

MULTIPLE LINEAR REGRESSION

- ❖ Multiple Linear Regression (MLR) is an extension of Simple Linear Regression that predicts a dependent variable (Y) using multiple independent variables ($X_1, X_2, X_3, \dots X_n$).
- ❖ It helps understand the relationship between multiple predictors and the target variable.

Applications of Multiple Linear Regression (MLR)

❖ Agriculture & Crop Yield Prediction

- ❖ Predicts crop yield based on rainfall, soil quality, fertilizer use, and temperature.
- ❖ Helps optimize resource allocation for better productivity.

❖ Real Estate Price Prediction

- ❖ Estimates house prices using size, number of bedrooms, location, and property age.
- ❖ Helps real estate agencies and banks in property valuation.

❖ Healthcare & Disease Prediction

- ❖ Predicts disease risk (e.g., heart disease, diabetes, or prostate cancer) based on age, cholesterol, BMI, and lifestyle.
- ❖ Used in medical diagnosis and treatment planning

- ❖ For example, 'area' of the house is useful to predict the house price.
- ❖ But in many practical cases, house price' does not dependent only on area. The price of a house can be decided depending on various factors like 'area', the number of bedrooms', the age of the house', etc.
- ❖ These variables are called independent variables and they are useful to predict the price of the house. When we use multiple (more than 1) variables in Linear Regression model, it is called Multiple Linear Regression Model'.
- ❖ **In Multiple Linear Regression, the target variable (y) value can be calculated based on several independent variables (x1, x2, x3, ...). So, the equation used by this model will be in the form of:**

$$y = m_1x_1 + m_2x_2 + m_3x_3 + b$$

- ❖ Here, y is called the dependent variable or target variable. x1, x2, x3, .. are called independent variables or features. m1, m2, m3, .. are called quotients associated with the dependent variables. b is called intercept.

- ❖ The relationship between y and x_1 should be linear. That means it can be shown in the form of a straight line. Similarly, the relationship between y and x_2 should be linear. Also, the relationship between y and x_3 should be linear.
- ❖ As usual, when Multiple Linear Regression model is applied on data, there will be certain deviations between the predicted values and original values which can be calculated through r squared value.
- ❖ The **r squared value** should be between 0 and 1. If it is nearer to 0, then the model is not performing well. If it is nearer to 1, then the model is doing well and accuracy level is high.
- ❖ Let us apply Multiple Linear Regression model on the house prices data. We will first look at the data and then understand how to use the model on the data.

Develop a Python program utilizing Multiple Linear Regression to predict home prices in Monroe Township, NJ (USA). Given the attributes of square footage area, number of bedrooms, and age of the home, the program should predict the prices for the following properties:

- a. 3000 sqft area, 3 bedrooms, 40 years old**
- b. 2500 sqft area, 4 bedrooms, 5 years old**

Solution

The dataset: homeprices.csv

- ❖ This dataset represents home prices in Monroe Township, New Jersey, USA. This dataset is a sample dataset that contains only 6 rows and 4 columns.
- ❖ The columns are the area of the house in square foot, the number of bed rooms, age of the house in months and price of the house in dollars. This is shown in Figure:

area	bedrooms	age	price	
2600	3	20	550000	
3000	4	15	565000	
3200		18	610000	
3600	3	30	595000	
4000	5	8	760000	
4400	5	8	795000	

- ❖ We are supposed to calculate the home price' depending on the 'area', bedrooms' and age columns. So, the Multiple Linear Regression model uses the following formula:

$$y = m_1x_1 + m_2x_2 + m_3x_3 + b$$

$$\text{price} = m_1 * \text{area} + m_2 * \text{bedrooms} + m_3 * \text{age} + b$$

- ❖ Please observe that this dataset has a missing value in bedrooms' column. So, first of all let us clean the data or make the data ready for the model. For this purpose, we have to either delete that row that contains the missing value or substitute appropriate value in that place.

Import the Libraries

1. `import pandas as pd`
2. `import seaborn as sns`
3. `from sklearn.linear_model import LinearRegression`
4. `import matplotlib.pyplot as plt`

Load the dataset into dataframe

1. `df=pd.read_csv('homeprices.csv')`
2. `df`

	area	bedrooms	age	price
0	2600	3.0	20	550000
1	3000	4.0	15	565000
2	3200	NaN	18	610000
3	3600	3.0	30	595000
4	4000	5.0	8	760000
5	4400	5.0	8	795000

#find out any missing values in the dataset

1. `df.isnull().sum()`

```
area      0  
bedrooms  1  
age       0  
price     0  
dtype: int64
```

first let us find the median value for 'bedrooms' column as:

1. `df.bedrooms.median()`

4.0

#fill the missing data (NaN columns) with this median value

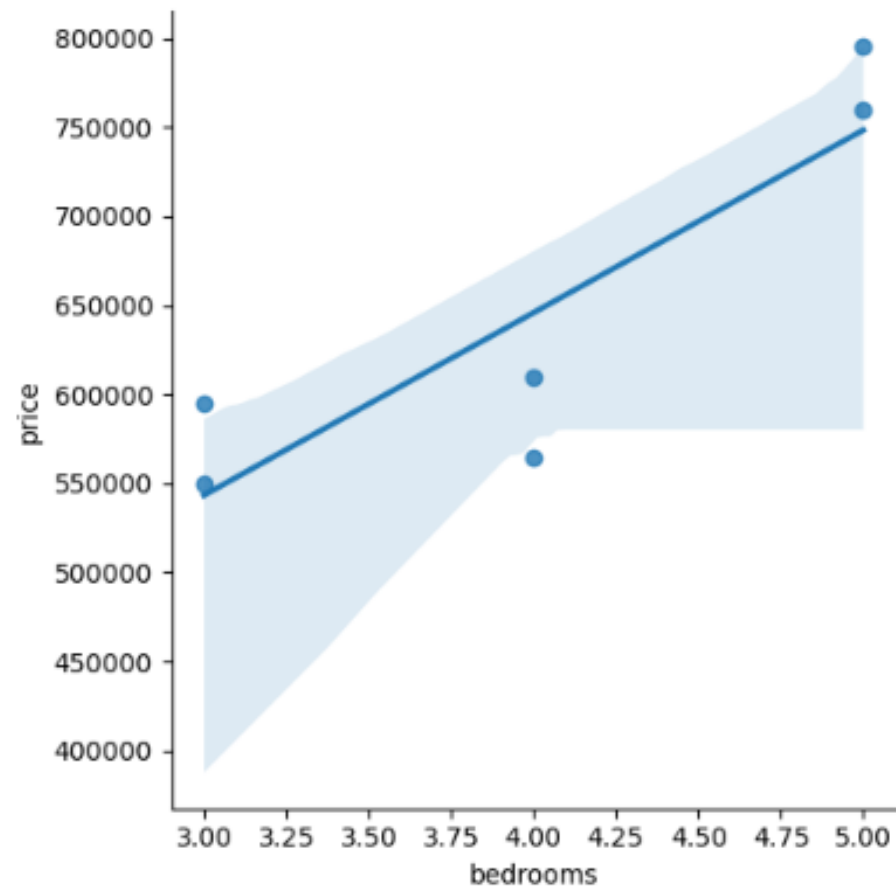
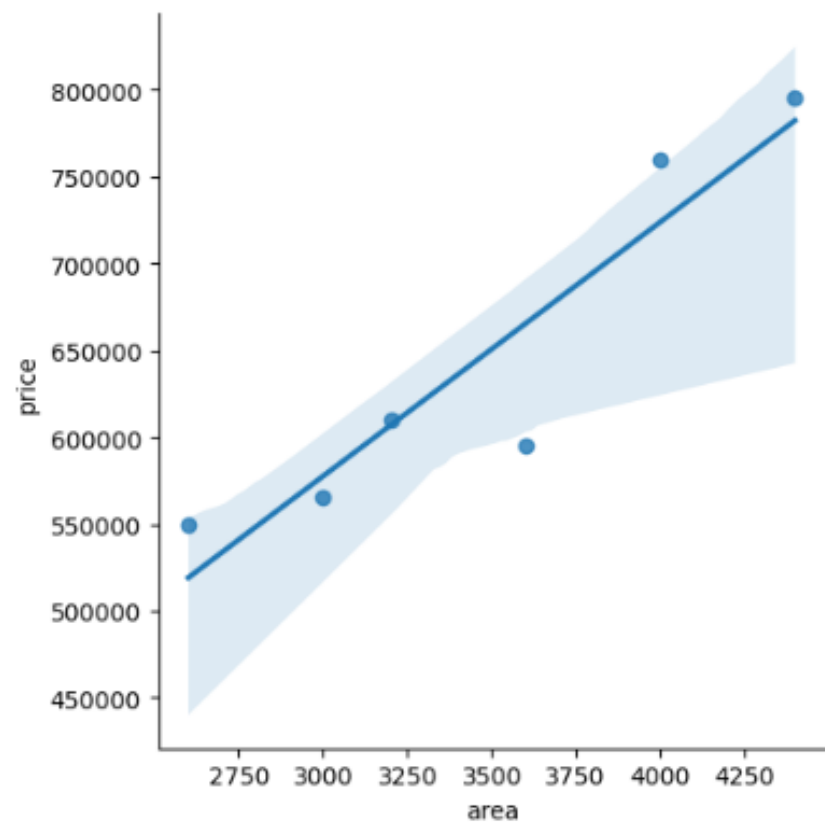
1. `df.bedrooms=df.bedrooms.fillna(med)`
2. `df`

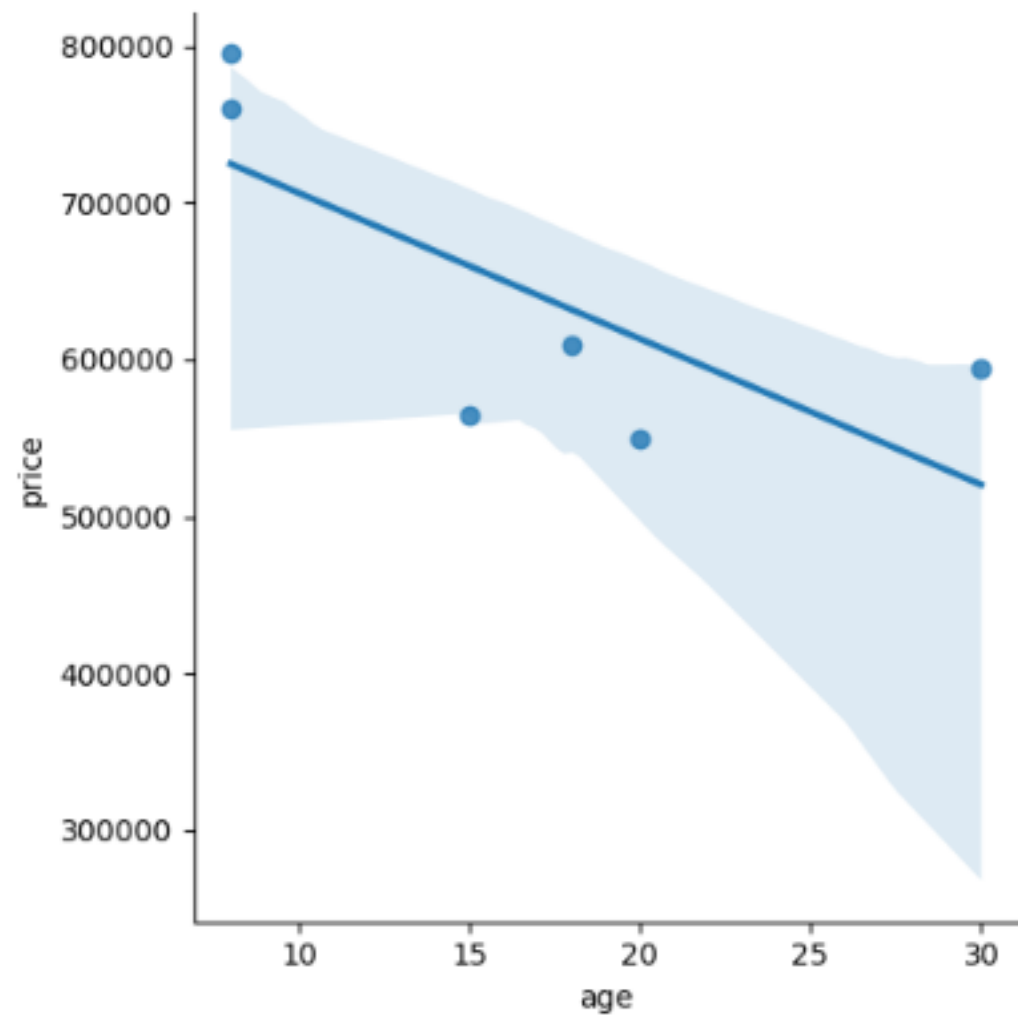
	area	bedrooms	age	price
0	2600	3.0	20	550000
1	3000	4.0	15	565000
2	3200	4.0	18	610000
3	3600	3.0	30	595000
4	4000	5.0	8	760000
5	4400	5.0	8	795000

represent the relationship between independent and dependent vars are, bedrooms and age are #indepndent vars and price is dependent

1. `sns.lmplot(x='area', y='price',data=df)`
2. `sns.lmplot(x='bedrooms', y='price',data=df)`
3. `sns.lmplot(x='age', y='price',data=df)`

<seaborn.axisgrid.FacetGrid at 0x1ebb8bc4f40>





#create linear regression model with multiple variables

#take the independent vars first and take dependent var next.

1. `from sklearn.linear_model import LinearRegression`
2. `reg=LinearRegression()`

#fitting means training

1. `reg.fit(df[['area','bedrooms','age']], df['price'])`

#print coefficients i.e m1, m2, m3 values

1. `reg.coef_`

```
array([ 142.895644 , -48591.66405516, -8529.30115951])
```

#intercept

```
1. reg.intercept_
```

```
485561.8928233979
```

#predict the price of 3000 sqft area, 3 bed rooms, 40 years old house

```
1. reg.predict([[3000,3,40]])
```

```
array([427301.78627387])
```

#predict the price of 2500 sqft area, 4 bed rooms, 5 years old house

```
1. reg.predict([[2500,4,5]])
```

```
array([605787.84080221])
```

Calculate R-squared value

1. `r_squared = reg.score(df[['area', 'bedrooms', 'age']], df['price'])`
2. `print(f"R-squared value: {r_squared:.4f}")`

```
R-squared value: 0.9761
```

Aim of the Program

- ❖ The aim of this program is to predict home prices in Monroe Township, NJ using Multiple Linear Regression. The model is trained based on square footage, number of bedrooms, and age of the home to estimate property values accurately.

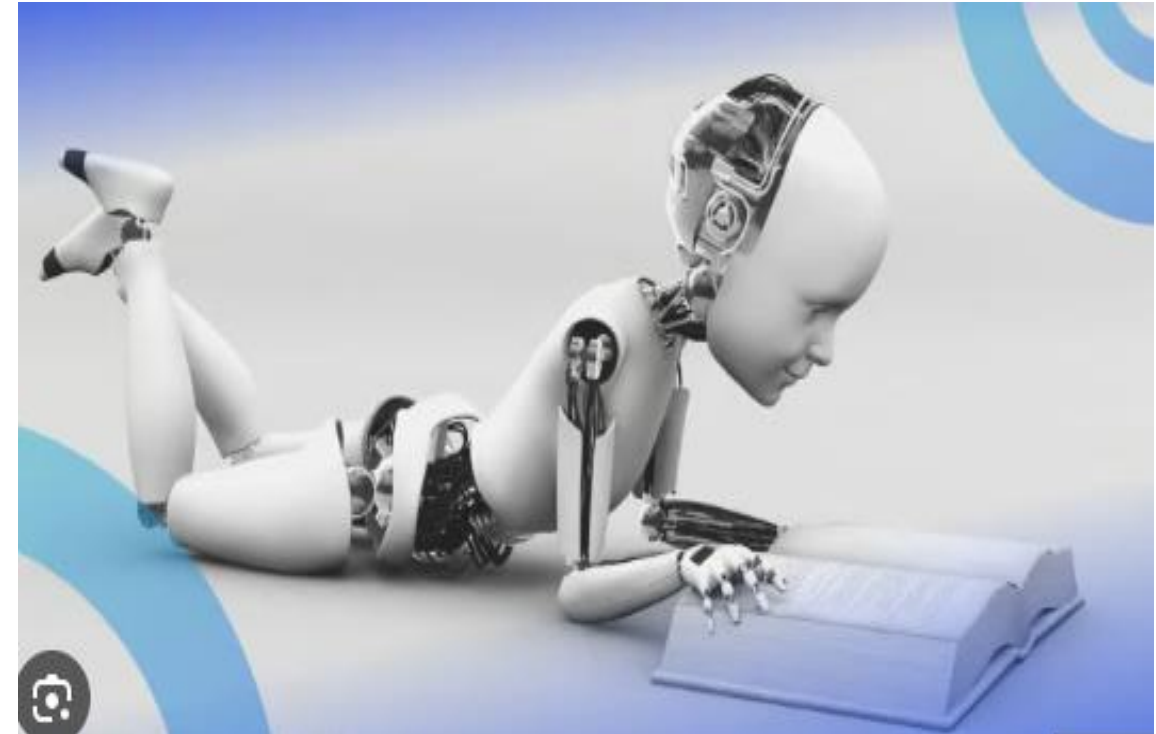
Conclusion

This program successfully utilizes Multiple Linear Regression to predict home prices in Monroe Township, NJ based on square footage, number of bedrooms, and age of the home. The model demonstrates how regression analysis can be effectively applied in real estate valuation, helping estimate property prices accurately.

Maximum Likelihood Estimation (MLE)

EM Algorithm in Machine Learning

- ❖ The EM algorithm is considered a latent variable model to find the local maximum likelihood parameters of a statistical model, proposed by **Arthur Dempster, Nan Laird, and Donald Rubin in 1977**.
- ❖ The EM (Expectation-Maximization) algorithm is one of the most commonly used terms in machine learning to obtain maximum likelihood estimates of variables that are sometimes observable and sometimes not.
- ❖ However, it is also applicable to unobserved data or sometimes called latent.
- ❖ It has various real-world applications in statistics, including obtaining the mode of the posterior marginal distribution of parameters in machine learning and data mining applications.



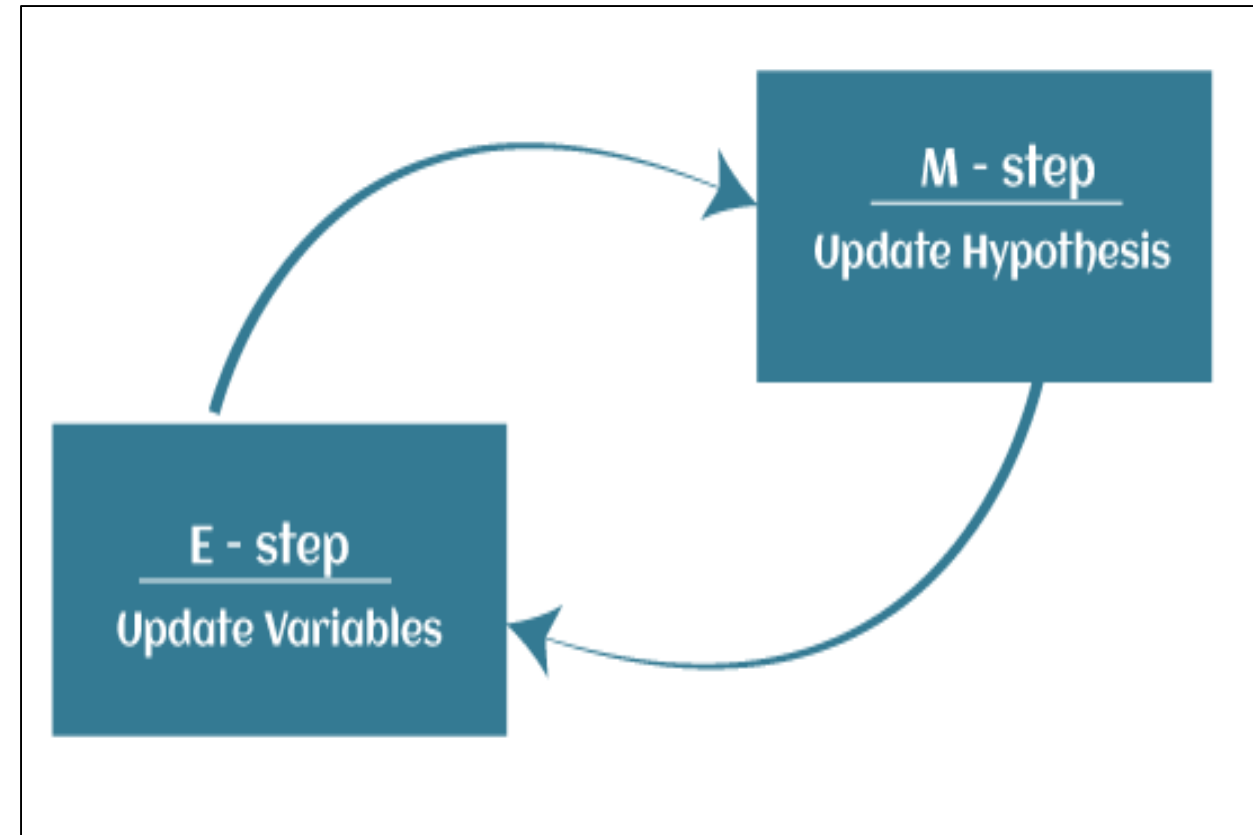
What is an EM algorithm?

- ❖ The Expectation-Maximization (EM) algorithm is defined as the combination of various unsupervised machine learning algorithms, which is used to determine the local maximum likelihood estimates (MLE) or maximum a posteriori estimates (MAP) for unobservable variables in statistical models.
- ❖ Further, it is a technique to find maximum likelihood estimation when the latent variables are present.
- ❖ It is also referred to as the latent variable model.
- ❖ A latent variable model consists of both observable and unobservable variables where observable can be predicted while unobserved are inferred from the observed variable.
- ❖ These unobservable variables are known as latent variables.



EM Algorithm

- ❖ The EM algorithm is the combination of various unsupervised ML algorithms, such as the k-means clustering algorithm.
- ❖ Being an iterative approach, it consists of two modes.
- ❖ In the first mode, we estimate the missing or latent variables. Hence it is referred to as the Expectation/estimation step (E-step).
- ❖ Further, the other mode is used to optimize the parameters of the models so that it can explain the data more clearly.
- ❖ The second mode is known as the maximization-step or M-step.



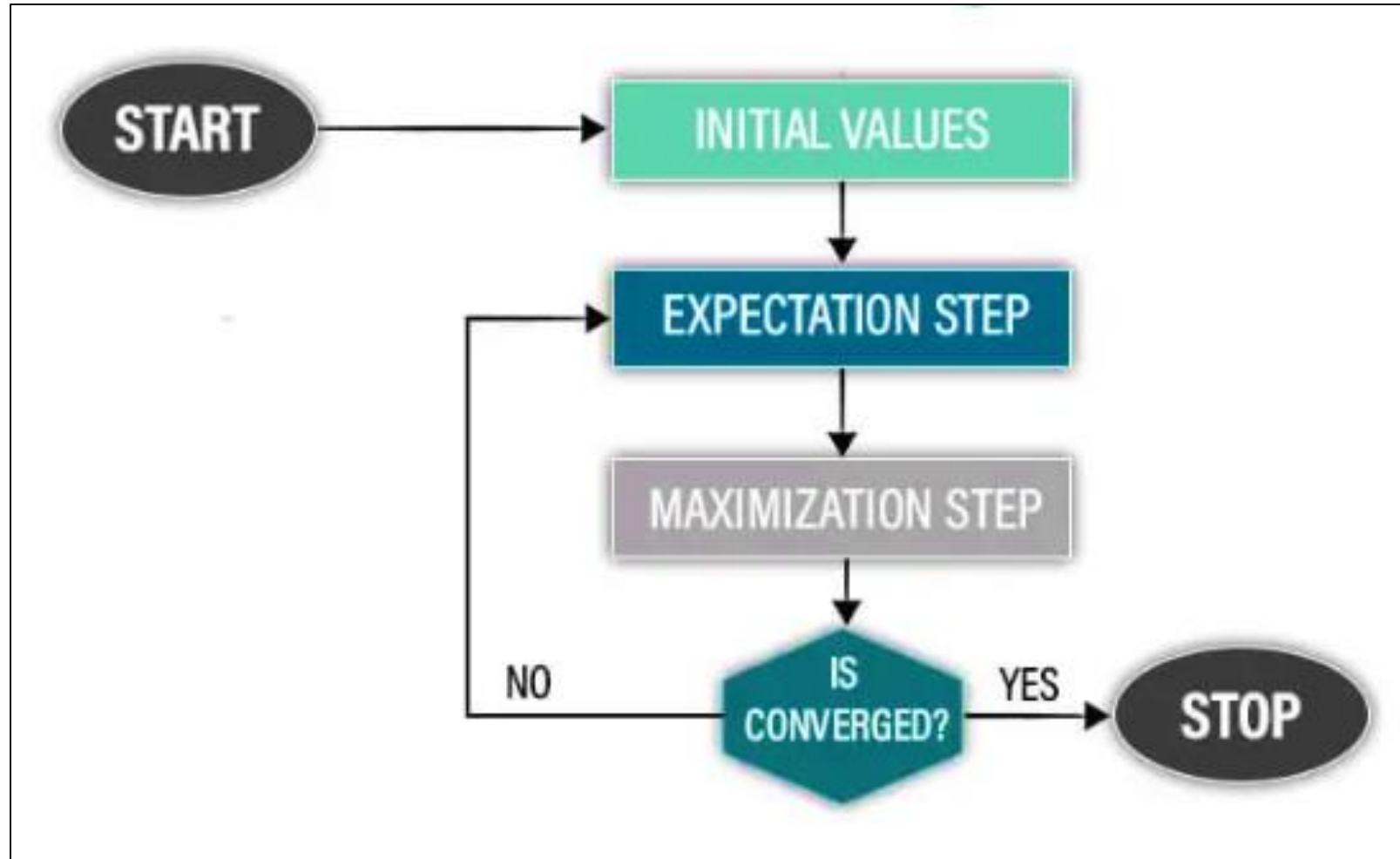
- ❖ **Expectation step (E - step):** It involves the estimation (guess) of all missing values in the dataset so that after completing this step, there should not be any missing value.
- ❖ **Maximization step (M - step):** This step involves the use of estimated data in the E-step and updating the parameters.
- ❖ Repeat E-step and M-step until the convergence of the values occurs.
- ❖ **The primary goal of the EM algorithm is to use the available observed data of the dataset to estimate the missing data of the latent variables and then use that data to update the values of the parameters in the M-step.**

What is Convergence in the EM algorithm?

- ❖ Convergence is defined as the specific situation in probability based on intuition, e.g., if there are two random variables that have very less difference in their probability, then they are known as converged.
- ❖ In other words, whenever the values of given variables are matched with each other, it is called convergence.

Steps in EM Algorithm

❖ The EM algorithm is completed mainly in 4 steps, which include Initialization Step, Expectation Step, Maximization Step, and convergence Step. These steps are explained as follows:



1st Step:

- ❖ The very first step is to initialize the parameter values. Further, the system is provided with incomplete observed data with the assumption that data is obtained from a specific model.

2nd Step:

- ❖ This step is known as Expectation or E-Step, which is used to estimate or guess the values of the missing or incomplete data using the observed data. Further, E-step primarily updates the variables.

3rd Step:

- ❖ This step is known as Maximization or M-step, where we use complete data obtained from the 2nd step to update the parameter values. Further, M-step primarily updates the hypothesis.

4th step:

- ❖ The last step is to check if the values of latent variables are converging or not. If it gets "yes", then stop the process; else, repeat the process from step 2 until the convergence occurs.

Applications of EM Algorithm:

1. Gaussian Mixture Model (GMMs):

- ❖ The EM algorithm is widely used in clustering algorithms to estimate data with similar characteristics and handle uncertainties.
- ❖ It calculates the Gaussian density, weight, shape, and location of each Gaussian components cluster by capturing clustering characteristics.
- ❖ For example, it is used in target marketing strategies to identify different groups of customers based on their purchasing behavior, such as product preference, amount spent, buying frequency, etc.

2. Hidden Markov Models (HMMs):

- ❖ The Hidden Markov Model is a probabilistic model mainly used for sequential data analysis where hidden states influence the observed data. EM estimates the parameters of HMMs, which are Emission probabilities and Transition probabilities.
- ❖ Used for speech recognition, bioinformatics, and finance.
- ❖ For example, used in speech recognition for identifying phonemes and acoustic features of audio signals, the EM algorithm can transcribe the verbal word into text.

3. Natural Language Processing (NLP):

- ❖ In natural language processing, practitioners utilize the Expectation-Maximization algorithm for diverse tasks, including text categorization, sentiment analysis, machine translation, topic modeling, and part-of-speech tagging.
- ❖ EM algorithm aids in extracting meaningful insights from text content by analyzing and learning its pattern.
- ❖ For example, NPL can efficiently identify sentiments in text by reviewing and classifying social media posts into positive, negative, or neutral.

4. Computer Vision:

- ❖ Computer vision employs the EM algorithm to recognize objects and reconstruct images. It enables us to extract valuable information from the visual data and image content.
- ❖ For example, the EM algorithm aids in reconstructing a three-dimensional image from a two-dimensional image, and these images can be any medical scans like CT or MRI, which can enhance visualization of any injuries or disease by reconstructing the process.

5. Health and Medical Industry:

- ❖ The EM algorithm has a valuable use in the health care and medical industry for improving patient treatment and care by aiding the detailed information for diagnosis by reconstructing and enhancing the diagnosed image quality.
- ❖ For example, the EM algorithm helps oncologists in cancer radiation therapy reconstruct a three-dimensional tumor image from a CT scan to target only the affected area and avoid any healthy tissue damage.

Advantages of EM algorithm

- ❖ It is very easy to implement the first two basic steps of the EM algorithm in various machine learning problems, which are E-step and M- step.
- ❖ It is mostly guaranteed that likelihood will enhance after each iteration.
- ❖ It often generates a solution for the M-step in the closed form.

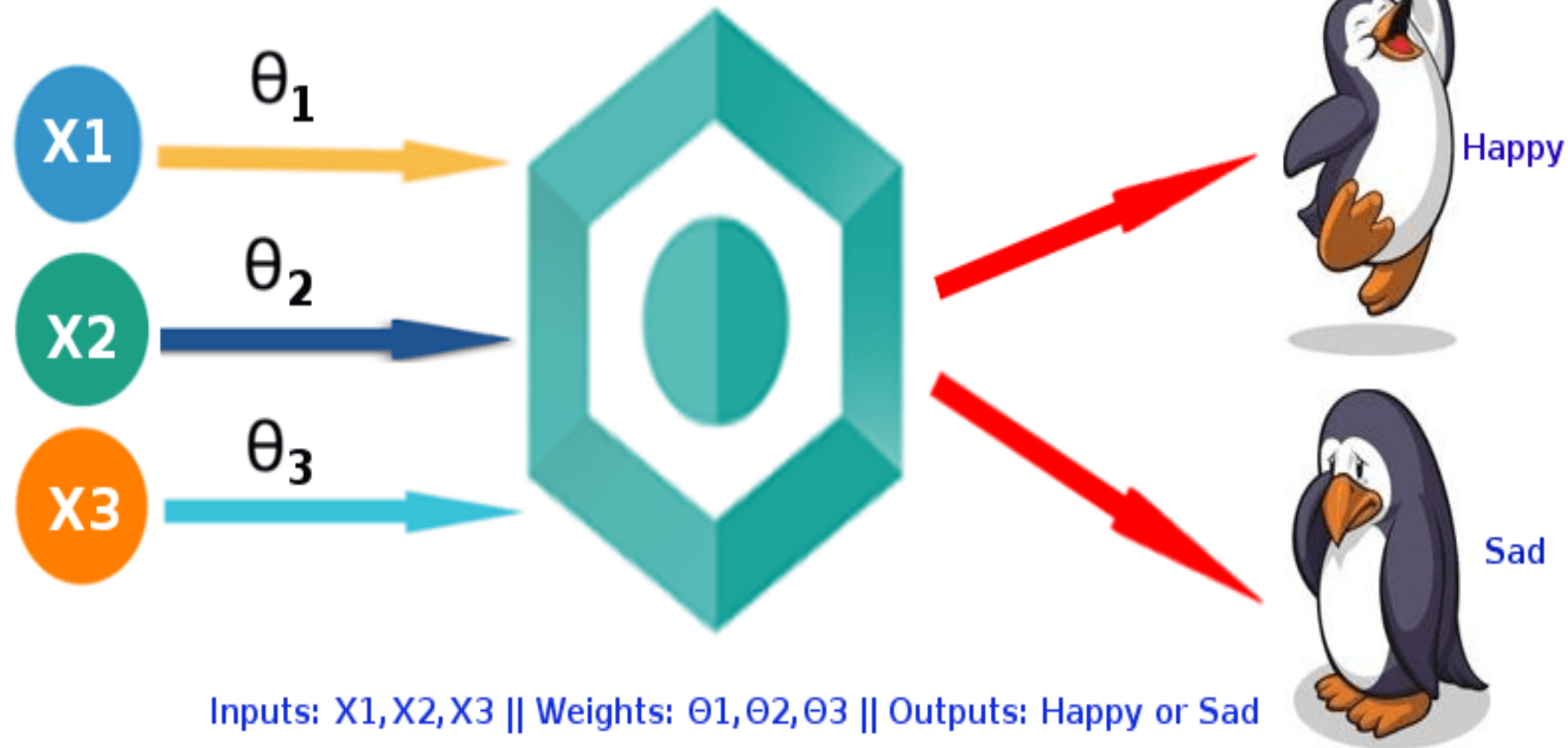
Logistic Regression Model

❖ **Linear regression models are fundamental in machine learning and statistics for predicting continuous outcomes based on input features.**

- ❖ Simple Linear Regression.
- ❖ Multiple Linear Regression.
- ❖ Polynomial Regression.
- ❖ Ridge Regression (L2 Regularization).
- ❖ Lasso Regression (L1 Regularization).
- ❖ Elastic Net Regression.
- ❖ Bayesian Linear Regression.
- ❖ Principal Component Regression (PCR).
- ❖ Partial Least Squares Regression (PLS Regression).
- ❖ Weighted Least Squares (WLS).
- ❖ Stepwise Regression.

- ❖ When we want to predict the categorical data, that means where the data is classified, we can think about Logistic Regression model. So, Logistic Regression is one of the techniques for classification.

Logistic Regression Model



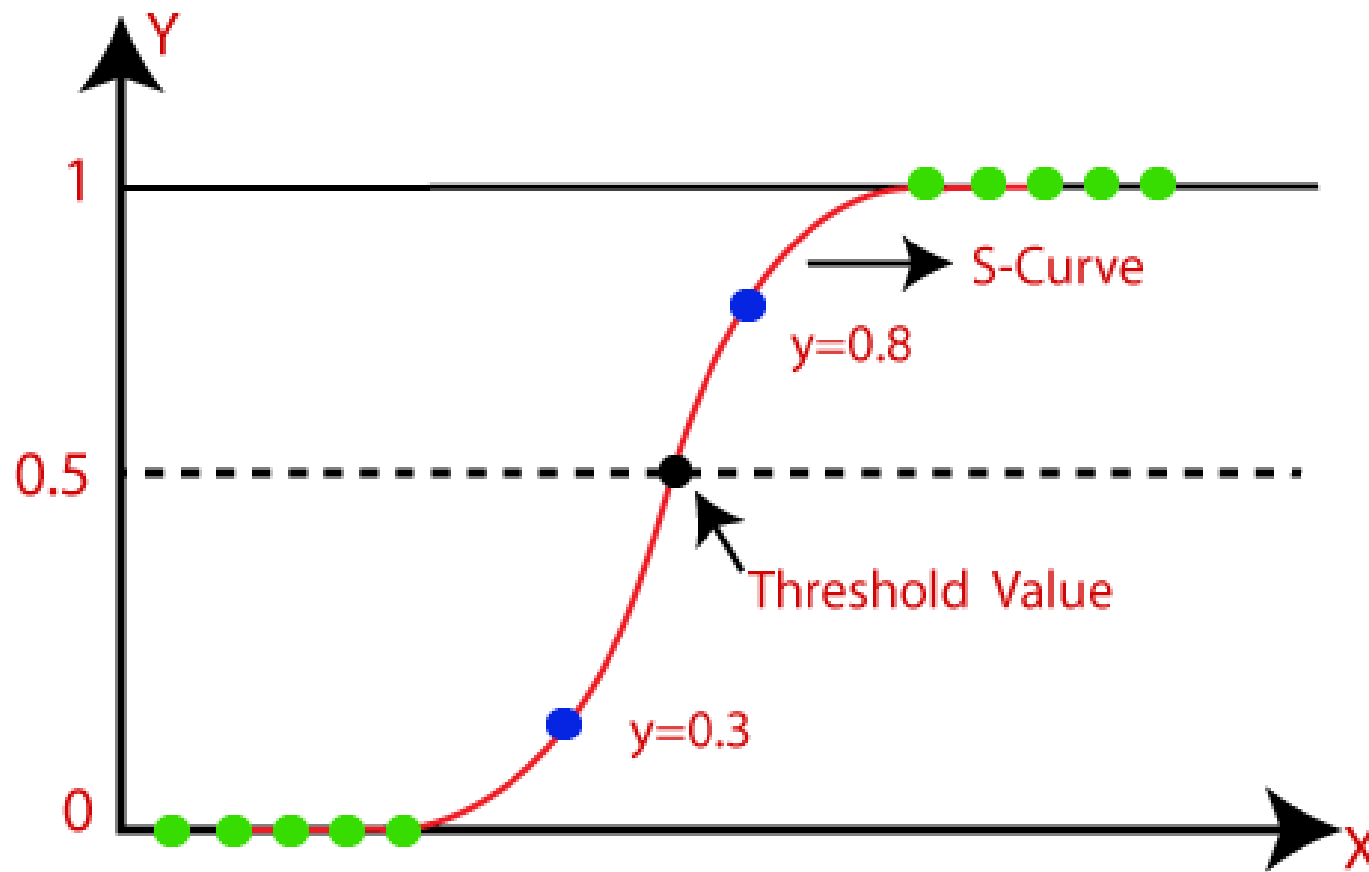
What Does Logistic Regression Mean?

- ❖ Logistic regression is a supervised learning algorithm used in machine learning to **predict the probability of a binary outcome**. A binary outcome is limited to one of two possible outcomes. Examples include yes/no, 0/1 and true/false.
- ❖ Logical regression is used in predictive modeling to analyze large datasets in **which one or more independent variables can determine an outcome**.
- ❖ The outcome is expressed as a **dichotomous variable** that has one of two possible outcomes.
- ❖ Essentially, logistic regression works by estimating the mathematical probability that an instance belongs to a specified class -- or not.
- ❖ Logistic regression uses something called the **Sigmoid function** to map predicted predictions and their probabilities.
- ❖ On a graph, if the estimated probability is **greater than a pre-defined** acceptance threshold, then the model will predict that the instance **belongs to that class**. If the estimated probability is **less than the pre-defined** threshold on the graph, then the model will predict the instance does not belong to the class.

- ❖ **The Sigmoid function performs the role of an activation function in machine learning which is used to add non-linearity in a machine learning model. Basically, the function determines which value to pass as output and what not to pass as output.**

Logistic Regression:

- ❖ Logistic regression is one of the most popular Machine learning algorithm that comes under Supervised Learning techniques.
- ❖ It can be used for **Classification as well as for Regression problems**, but mainly used for Classification problems.
- ❖ Logistic regression is used to predict the **categorical dependent variable** with the help of independent variables.
- ❖ The output of Logistic Regression problem can be only between the 0 and 1.
- ❖ Logistic regression can be used where the probabilities between two classes is required. Such as whether it will rain today or not, either 0 or 1, true or false etc.
- ❖ Logistic regression is based on the concept of Maximum Likelihood estimation. According to this estimation, the observed data should be most probable.
- ❖ In logistic regression, we pass the weighted sum of inputs through an activation function that can map values in between 0 and 1. Such activation function is known as **sigmoid function** and the curve obtained is called as **sigmoid curve or S-curve**. Consider the below image:



The equation for logistic regression is

$$\log \left[\frac{y}{1-y} \right] = \mathbf{b}_0 + \mathbf{b}_1 \mathbf{x}_1 + \mathbf{b}_2 \mathbf{x}_2 + \mathbf{b}_3 \mathbf{x}_3 + \cdots + \mathbf{b}_n \mathbf{x}_n$$

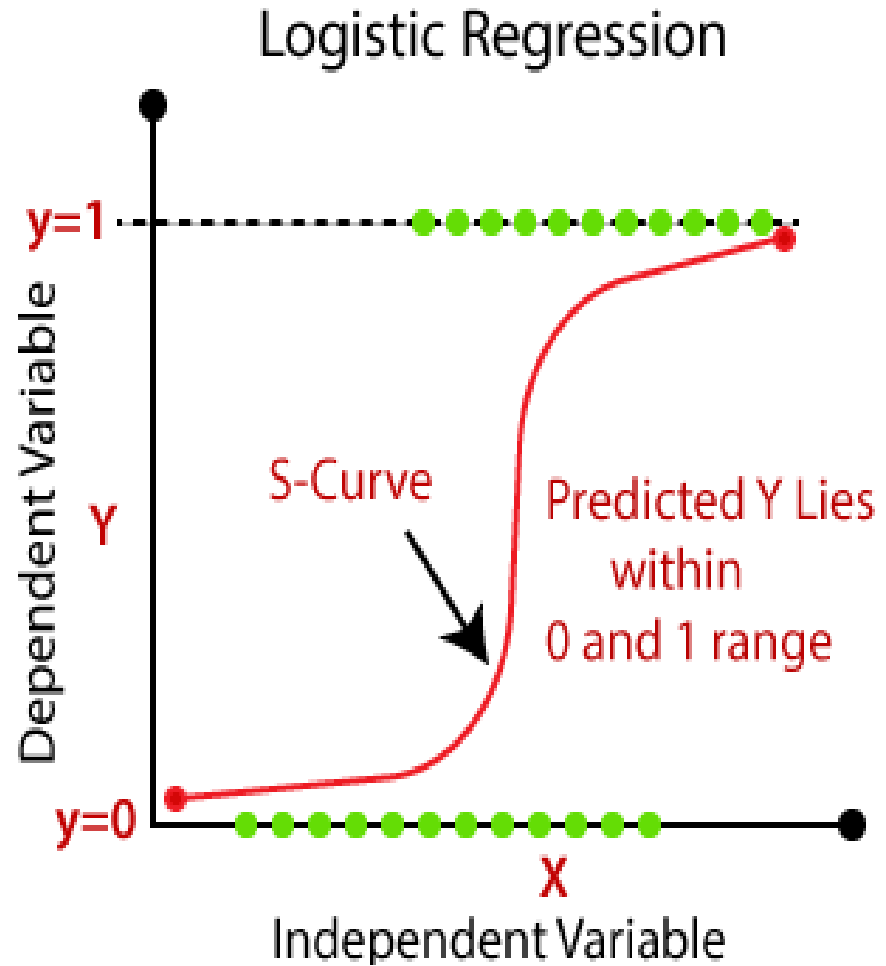
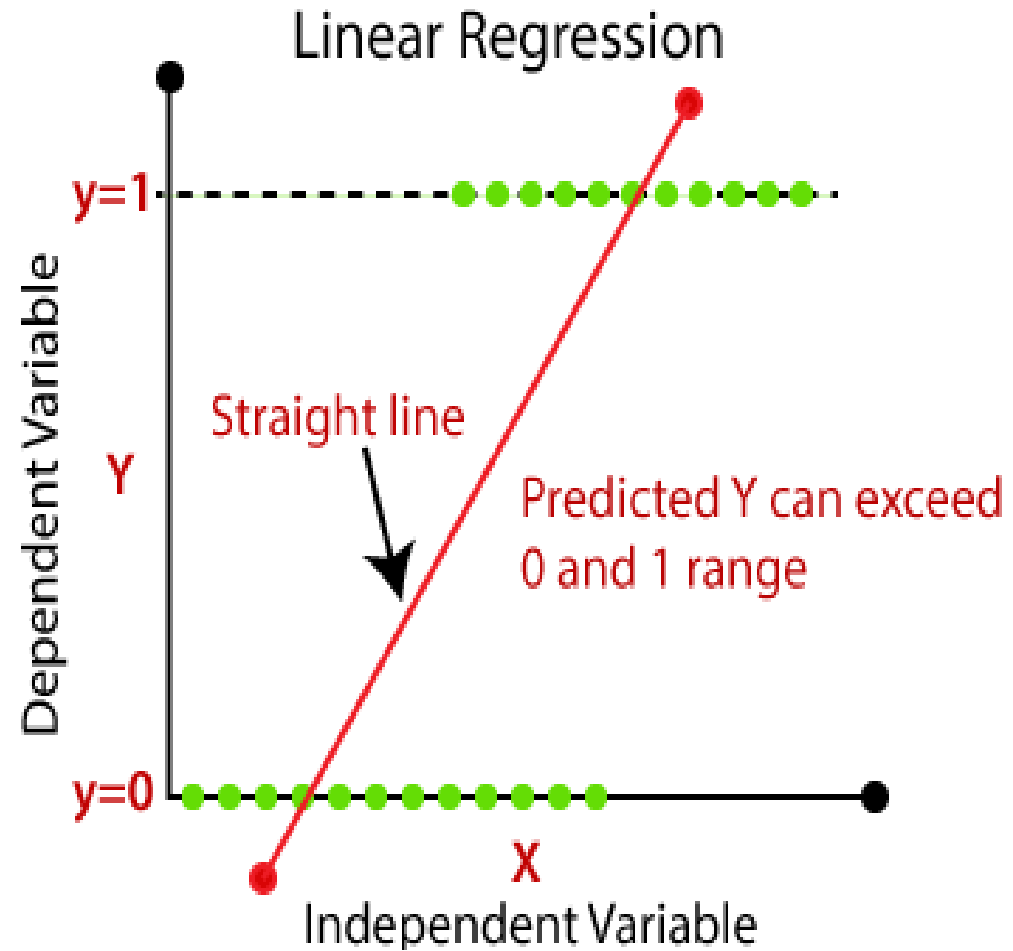
In statistics, there are three basic types of logistic regression:

- ❖ **Binary logistic regression** -- useful for predicting the relationship between a binary dependent variable (Y) and an independent variable (X).
- ❖ **Multinomial logistic regression** -- useful for making predictions when the dependent variable has two or more discrete outcomes and the order of the outcomes doesn't matter.
- ❖ **Ordinal logistic regression** -- useful for making predictions when the dependent variable has more than two discrete outcomes and the order of the outcomes has some significance.

Why do we use Logistic Regression rather than Linear Regression?

- ❖ Logistic Regression is used instead of Linear Regression for classification problems because:
- ❖ **Output Interpretation:** Logistic Regression predicts probabilities (values between 0 and 1), making it suitable for classification, whereas Linear Regression can produce values outside this range.
- ❖ **Decision Boundary:** It applies a sigmoid function to ensure a clear decision boundary, whereas Linear Regression does not provide distinct class separation.

The description of both the algorithms is given below along with difference table.



Classification Types

- ❖ When the categorical data shows only 2 possibilities, then it is called '**Binary Classification**'. For example,
 - ❖ Will the customer buy life insurance? 1. Yes, 2. No.
 - ❖ What is the gender of the person? 1. Male, 2. Female.
- ❖ When we have more than 2 possibilities, then it is called '**Multiclass Classification**'. In this the number of possibilities can be 3, 4 or even more. For example,
 - ❖ Which party a person is going to vote for? 1. BRS, 2. BJP, 3. Congress
 - ❖ What is this written digit? May be any digit from 0 to 9. Hence 10 possibilities.

Binary Classification using Logistic Regression

- ❖ In Binary classification, we have to predict one of the two classes or groups of data.
- ❖ For example, we are given age of the person and whether he bought insurance or not. Depending on age, we have to predict the person will buy insurance or not. That means the variable is : buying insurance and its possibilities are : 'Yes' or 'No'.

Program 1:

We are given data like age and bought_insurance.

Apply Logistic Regression Model and predict whether a person takes insurance or not based on his age.

#1 logistic regression - binary classification

1. import pandas as pd
2. import matplotlib.pyplot as plt

load the data into dataframe

1. df = pd.read_csv("C:\\Exp\\DA\\insurance_data.csv")
2. df

```
: df = pd.read_csv("C:\\Exp\\DA\\insurance_data.csv")  
df
```

```
:
```

	age	bought_insurance
0	22	0
1	25	0
2	47	1
3	52	0
4	46	1
5	56	1
6	55	0
7	60	1
8	62	1
9	61	1
10	18	0
11	28	0
12	27	0
13	29	0
14	49	1
15	55	1
16	25	1

retrieve the data

1. `x=df.iloc[:, :-1].values` #retrieve only 0th column
2. `x`
3. `y=df.iloc[:,1].values` #retrieve the 1st column
4. `y`

```
y=df.iloc[:,1].values #retrieve the 1st column  
y
```

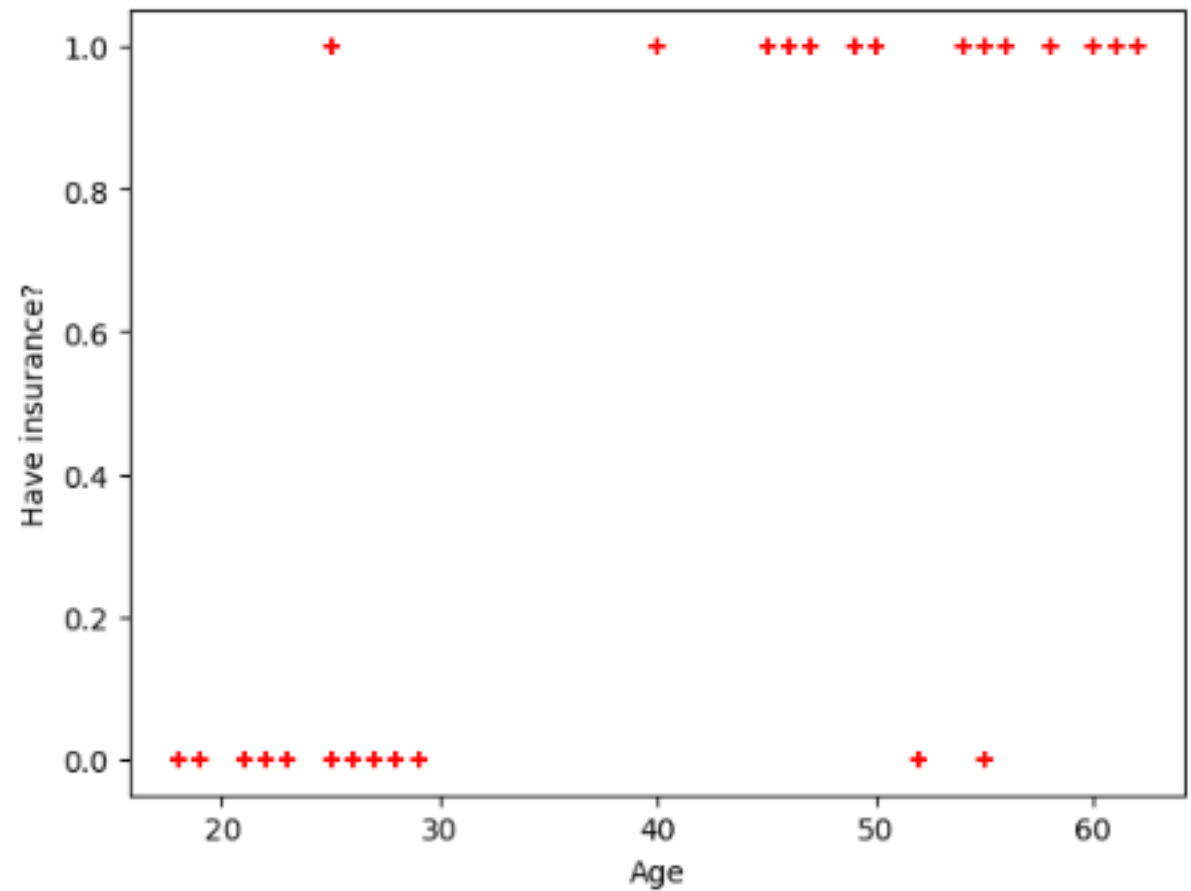
```
array([0, 0, 1, 0, 1, 1, 0, 1, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0,  
       1, 1, 1, 1, 0], dtype=int64)
```

```
x=df.iloc[:, :-1].values #re  
x
```

```
array([[22],  
       [25],  
       [47],  
       [52],  
       [46],  
       [56],  
       [55],  
       [60],  
       [62],  
       [61],  
       [18],  
       [28],  
       [27],  
       [29],  
       [49],  
       [55],  
       [25],  
       [58],  
       [19],  
       [18],  
       [21],  
       [26],  
       [40],  
       [45],  
       [50],  
       [54],  
       [23]], dtype=int64)
```

display the scatter plot to know how the datapoints are aligned

1. `plt.xlabel('Age')`
2. `plt.ylabel('Have insurance?')`
3. `plt.scatter(x,y, marker='+', color='red')`

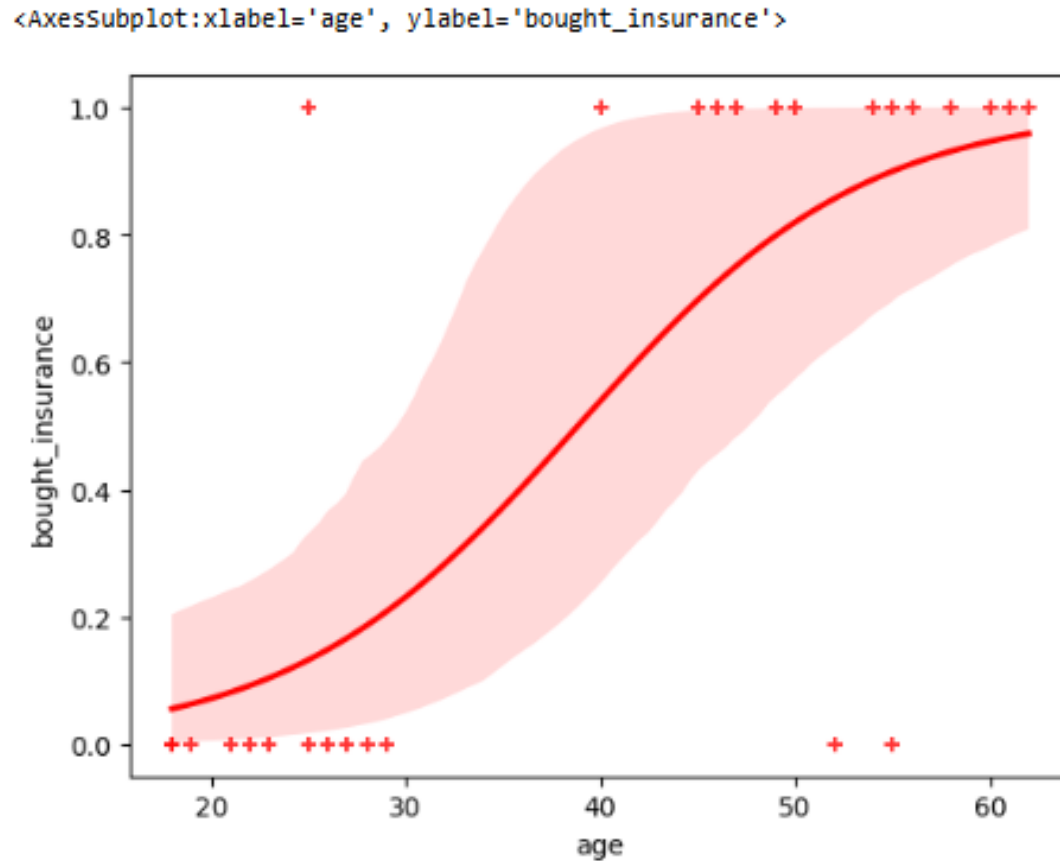


for the above data, we cannot use linear regression

#show the data as logistic regression plot

1. import seaborn as sns

2. sns.regplot(x='age', y='bought_insurance', data=df, logistic=True, marker='+', color='red')



create logistic regression model

1. from sklearn.linear_model import LogisticRegression
2. model = LogisticRegression()

train the model

1. model.fit(x,y)

find accuracy of the model

1. model.score(x, y)

Output:

```
: # find accuracy of the model  
model.score(x, y)
```

```
: 0.8888888888888888
```

predict if 56 years aged person will buy insurance or not

1. `model.predict([[56]])` # array([1]) yes

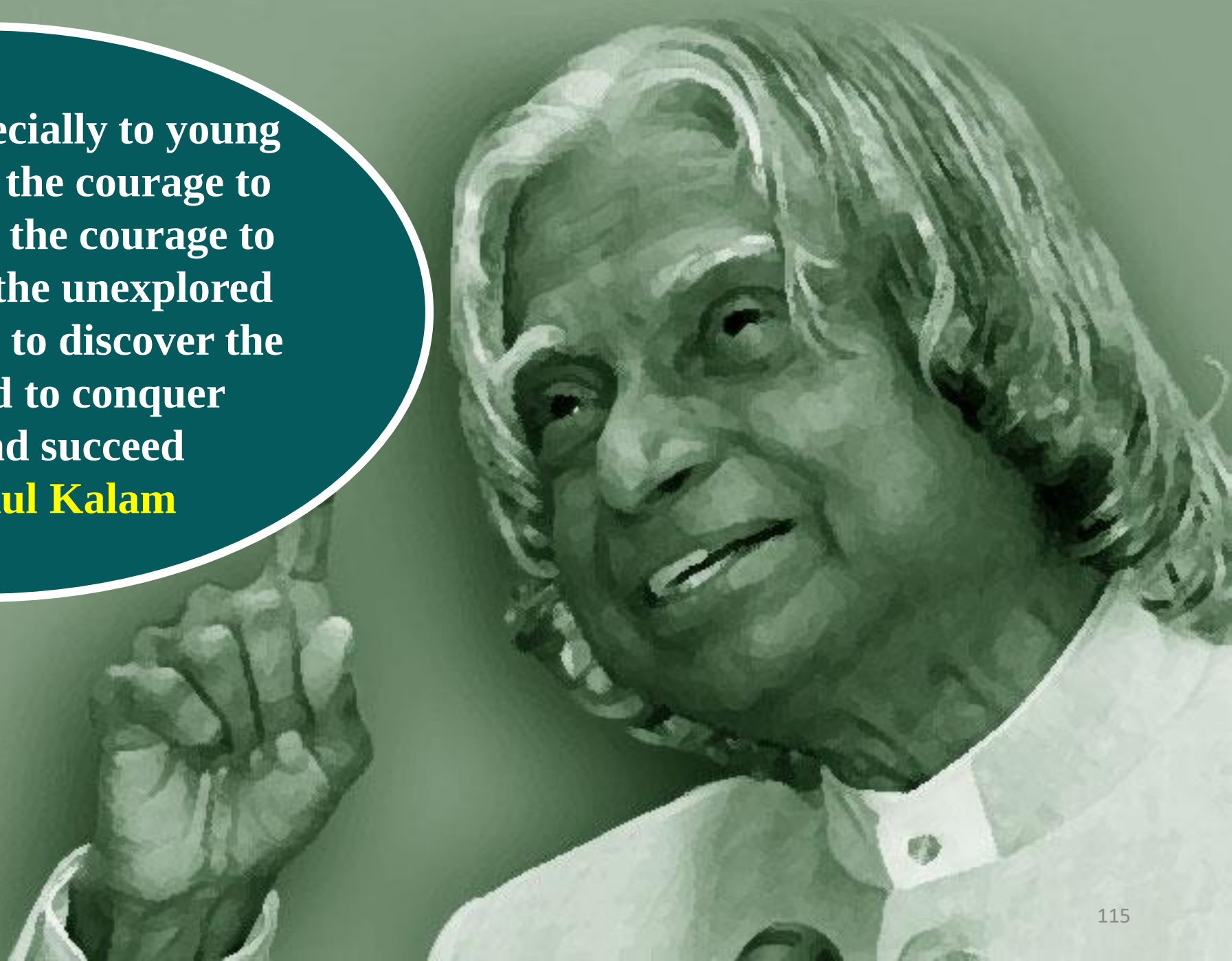
`array([1], dtype=int64)`

predict if 36 years aged person will buy insurance or not

1. `model.predict([[36]])` #array([0]) No

`array([0], dtype=int64)`

My message, especially to young people is to have the courage to think differently, the courage to invent, to travel the unexplored path, the courage to discover the impossible and to conquer problems and succeed
A. P. J. Abdul Kalam



A red pushpin is pinned to the top center of a yellow sticky note.

Thank
You!